UPSHIFT 2025

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1 Data description

1.1 Data Source

The data used in this report was taken from an experiment run in the TerraX Cube hypobaric chambers in Bolzano, IT in November 2025 on *Arabidopsis thaliana* individuals. Data collected by Marta De Giuli, Bouchra El Omari, and Emily Mason and compiled by Marta De Giuli in 2026.

1.2 Description of the data

Description of the data and the question(s).

- 4 chambers for a 2 factorial design :
  - 2 pairs at the same elevation/pressure
  - 2 pairs at the same D/pressure corrected VPD
- In each chamber we have 30 pots each containing 3-4 individuals, kept at 100% soil saturation.
- We have a total of 120 pots across all chambers.
- We have temporal data for select measurements. A selection of non-destructive measurements were taken halfway through and at the end of the experiment, while other - namely the more destructive ones were taken only at the end of the two week exposure to experimental conditions.

- Our main goal is to understand *if* and *how* pressure impacts ecophysiological parameters of *Arabidopsis thaliana* and how pressure interacts with humidity. The goal is to disentangle the effect of pressure and humidity to better understand the stressors high alpine plants are subjected to.

Covariates:

- Chlorophyll fluorescence measurements, as taken with a FluorCAM 600 from leaf discs:
  - PSII (Y), NPQ, 1-qL
- Photosynthetic measurements, as taken with a WALZ device:
  - Net carbon assimilation, gas exchange (stomatal conductance), evapotranspiration
  - and calculated: Ci:Ca
- Metabolitic measurements
  - C:N ratios, d13C values
  - chlorophyll content, carotene content, chlorophyll a:b ratio
- Biometric measurements including:
  - plant height, rosette diameter and leaf number
  - fresh weight, turgid weight, dry weight, and calculated RWC

2 Analysis

Before starting with the deeper analyses, it is important to check that our treatments *pressure* x D (normalized VPD) worked as expected. To do so, we placed two beakers in each chamber and measured evaporation throughout the experiment.

2.1 Evaporation

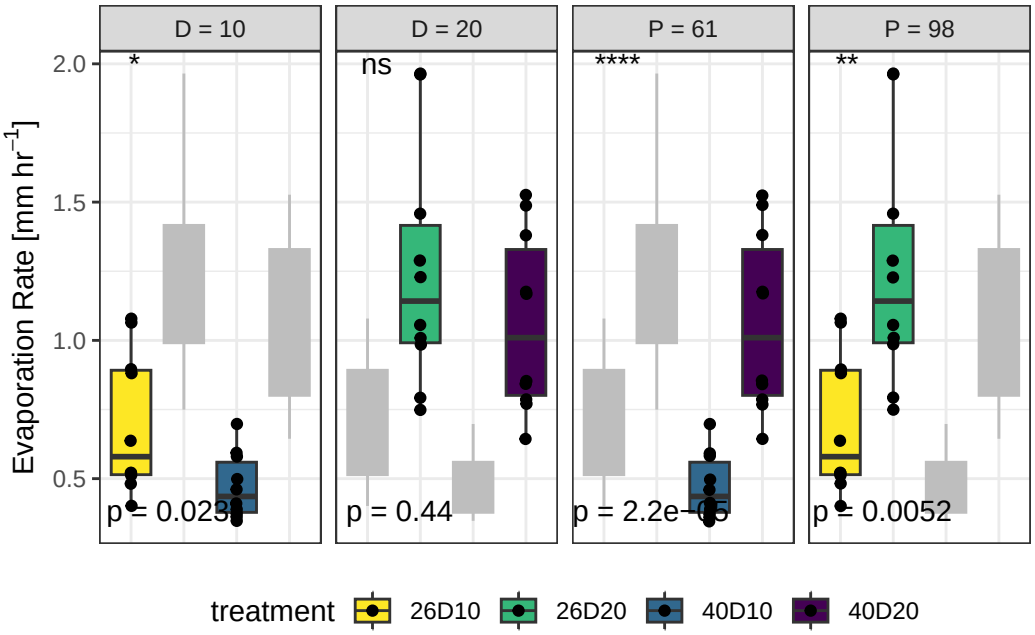


Figure 1: Evaporation rate of beakers placed in each chamber - aggregated by treatment. Comparisons are made between treatments with the same D value (D = 10 and D = 20) and pressure (P = 61 kPa and P = 98 kPa.

As expected, we observe a strong statistical difference between humidities at the same pressure (panels P= 61 and P = 98), with chambers at D = 20 showing higher evaporation. Slight differences are also present between elevations at the same humidity, with chambers at lower pressure generally showing a lower evaporation rate (specifically D = 10, where we observe a significant decrease, p = 0.023). Thus we will move forward with our analyses.

2.2 Chamber parameters

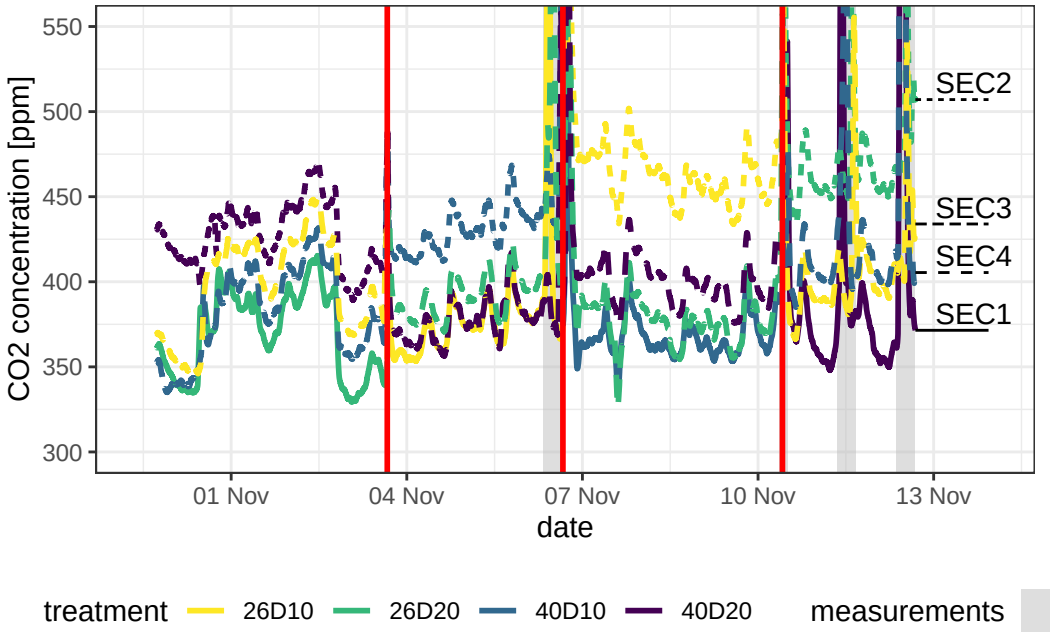


Figure 2: Concentration of CO_2 in ppm of each chamber, the data is a 30min running mean of the raw (1min resolution) data. Colors indicate treatment present in each chamber, red shaded area indicate periods when measurements were taking place (expect higher ppm CO_2 due to human presence), line type indicates the chamber, and red vertical bars indicate rotations of treatments between the chambers.

There is variability in the CO_2 concentration between the chambers, with SEC2 having a consistently higher concentration of CO_2 than the others. Each treatment was exposed to each chamber, and photosynthetic parameters were taken using the in-situ ppm of CO_2 (instantaneous reading).

2.3 Chlorophyll Fluorescence

2.3.1 Aggregating humidities (and years?)

In previous discussions, we have spoken about the fact that humidity seems to have a much smaller impact on chlorophyll fluorescence parameters than pressure. To check whether this is true, we can compare four nested model:

$$response \sim irradiance + pressure \tag{1}$$
$$response \sim irradiance + pressure + time \tag{2}$$
$$response \sim irradiance + pressure + D + time \tag{3}$$
$$response \sim irradiance + pressure + D + pressure * D + time \tag{4}$$

This will also enable us to understand whether there are general *interacting effects* of pressure and humidity. Here we look at the case for the yield of PSII (see Section 3.1 for the other response terms).

```
# Analysis of Variance Table
#
# Model 1: QY ~ irr + pressure
# Model 2: QY ~ irr + pressure + time
# Model 3: QY ~ irr + pressure + D + time
# Model 4: QY ~ irr + pressure + D + pressure * D + time
#   Res.Df    RSS Df Sum of Sq      F Pr(>F)
# 1    597 10.833
# 2    596 10.831  1 0.0022589 0.1239 0.7249
# 3    595 10.827  1 0.0034762 0.1907 0.6625
# 4    594 10.826  1 0.0010610 0.0582 0.8094
```

Since the explanatory power of the models doesn't decrease with the removal of the interaction + humidity terms, it seems that the best model is the one that doesn't include humidity at all. This indicates that we can make our comparisons looking simply at the difference between pressure treatments, aggregating our observations by pressure. Based on these results, it would be statistically sound not to differentiate by measurement time as no *significant* change in explanatory power is seen. This tends towards pseudo-replication though, and would likely become problematic to defend to a potential reviewer.

Seeing the result indicating that humidity doesn't add explicative value to the model, it could be possible to aggregate last experiment's data to this year's. To test whether it is statistically reasonable to do, we compare whether a model including year as a fixed effect, performs better to explain a combined FY 2024 and FY 2025 dataset. The two compared models will be:

$$psii \sim irradiance + pressure + time \tag{5}$$

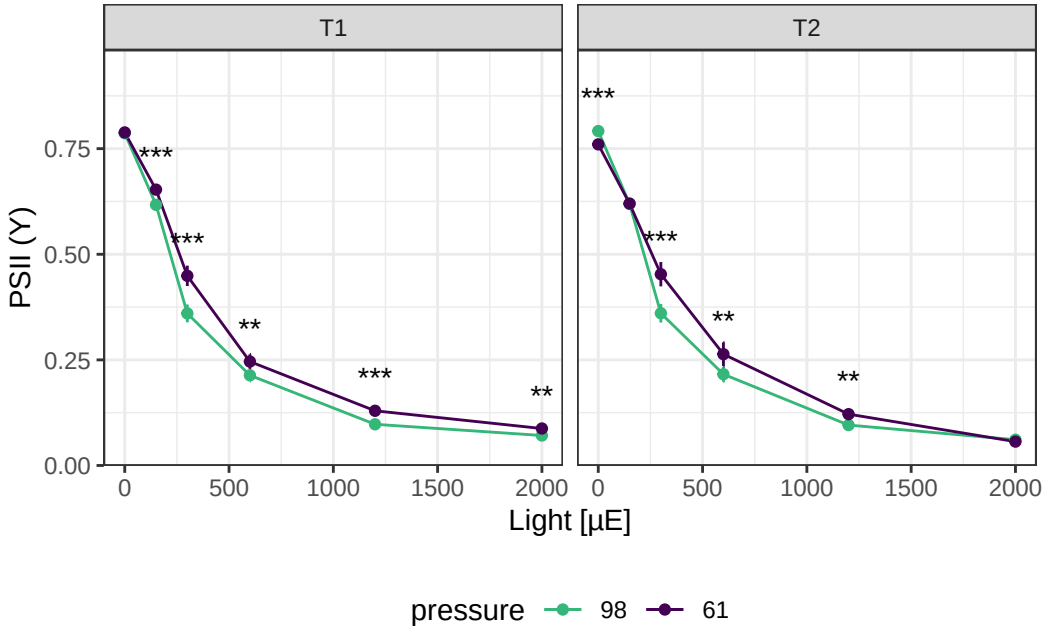
$$psii \sim irradiance + pressure + time + year \tag{6}$$

```
# Analysis of Variance Table
#
# Model 1: psii ~ irr + pressure + time
# Model 2: psii ~ irr + pressure + time + year
#   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
# 1   1120 37.095
# 2   1119 33.885  1    3.2103 106.02 < 2.2e-16 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

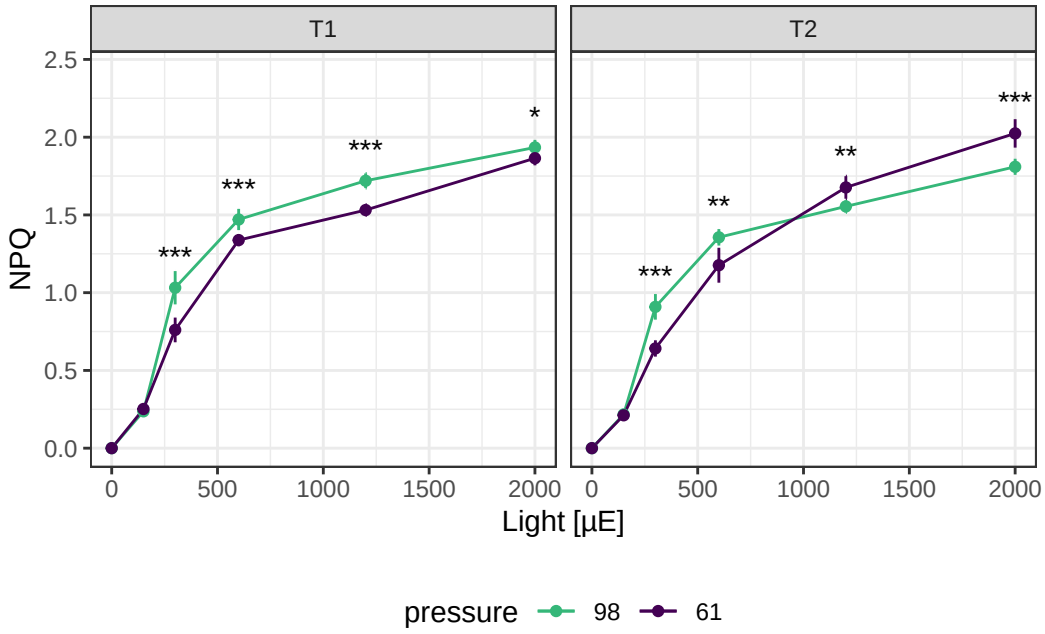
The model with year as fixed effect performs significantly better, explaining the variance best, indicating that year is needed to explain the variance in the aggregated data. This

means that we cannot aggregate results from the separate years, without considering year as a fundamental parameter. We will only consider data FY 2025, aggregating as mentioned above by pressure.

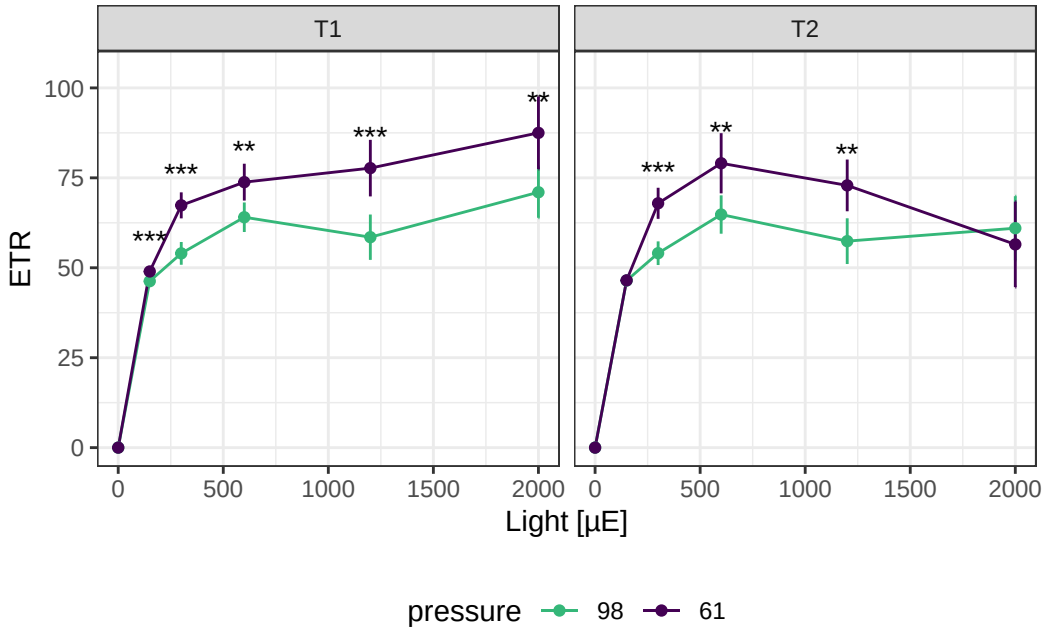
2.3.2 PSII(Y)



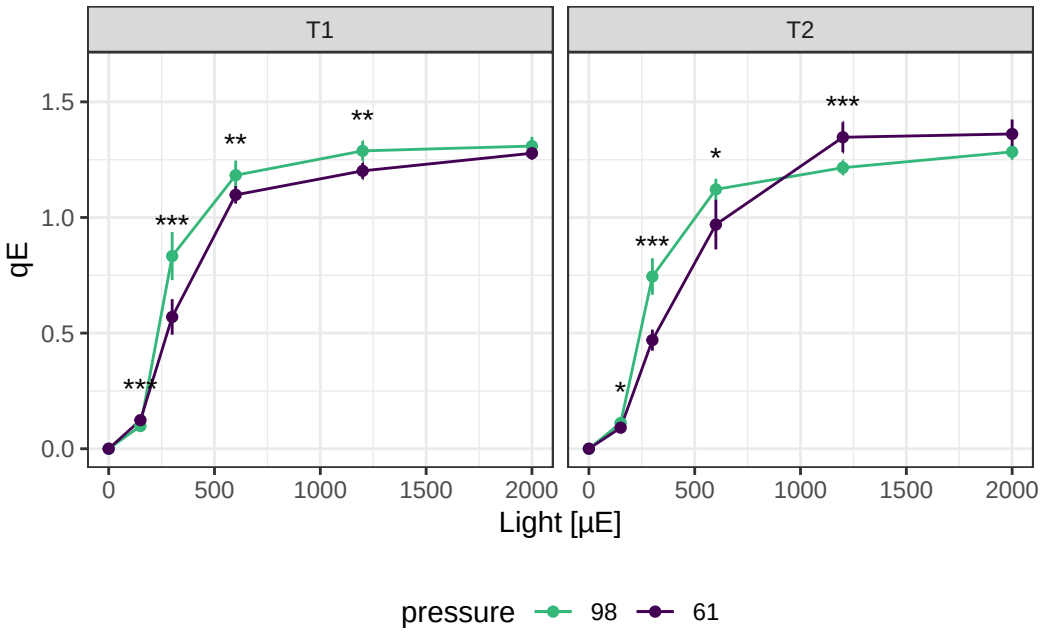
2.3.3 NPQ



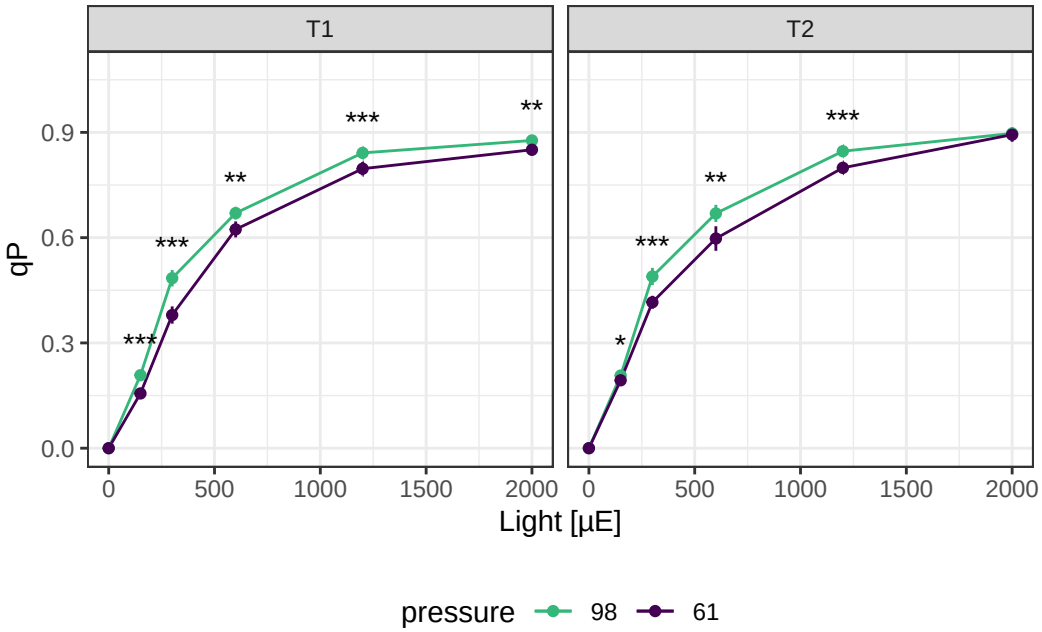
2.3.4 ETR



2.3.5 qE



2.3.6 qP



We see aligned trends across all the chlorophyll fluorescence parameters aggregated by pressure: *Arabidopsis thaliana* plants exposed to low pressure had a higher quantum yield

of PSII, lower NPQ, a higher electron transport rate and lower qP and qE. These results are consistent in indicating a higher PSII operating efficiency, and a reduced reliance on photoprotective energy dissipation mechanisms in plants grown at low pressure.

The treatment level (Section 3.3) and temporal level (Section 3.4) results are presented in the supplementary materials.

2.4 Photosynthesis

We know that these measurements depend on temperature as well and in Section 3.5 we provide the leaf temperature data - which varies from 25.0°C to 24.55°C across treatments.

To better understand the interaction of our manipulated parameters, it would be wise to look at how a similar model scheme compares for our photosynthetic response parameters.

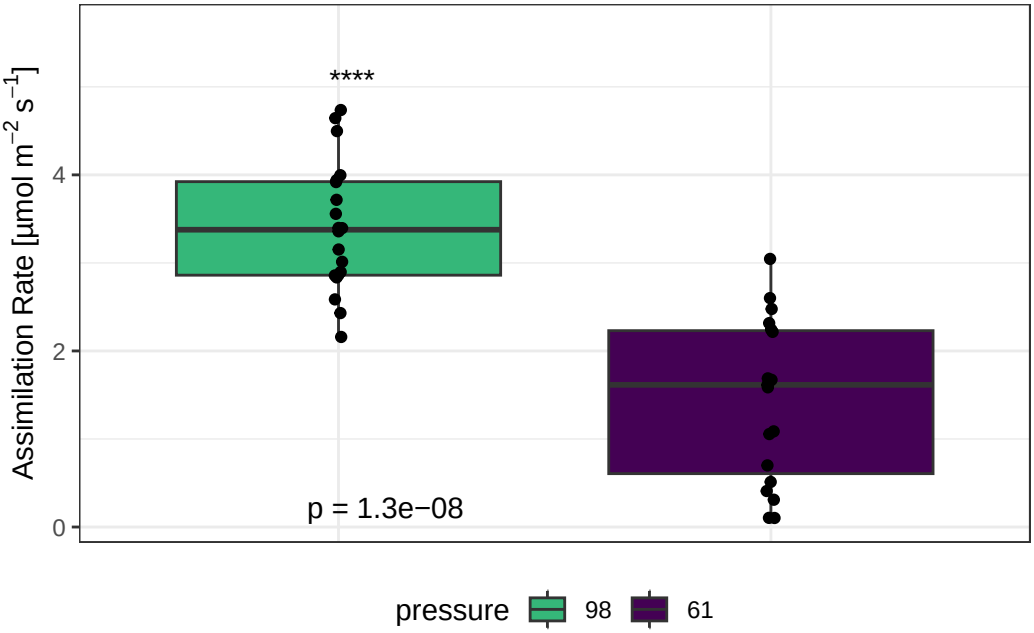
2.4.1 Assimilation Rate

For assimilation rate, we will compare the following three nested models:

$$assimilation \sim pressure \tag{7}$$
$$assimilation \sim pressure + D \tag{8}$$
$$assimilation \sim pressure * D \tag{9}$$

```
# Analysis of Variance Table
#
# Model 1: A ~ pressure
# Model 2: A ~ pressure + D
# Model 3: A ~ pressure + D + pressure * D
#   Res.Df    RSS Df Sum of Sq      F Pr(>F)
# 1     37 24.917
# 2     36 24.868  1   0.04934 0.0760 0.7844
# 3     35 22.712  1   2.15588 3.3223 0.0769 .
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# Analysis of Variance Table
#
# Response: A
#            Df Sum Sq Mean Sq F value    Pr(>F)
# pressure    1 37.251  37.251 57.4044 6.996e-09 ***
# D           1  0.049   0.049  0.0760    0.7844
# pressure:D  1  2.156   2.156  3.3223    0.0769 .
# Residuals  35 22.712   0.649
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Comparing the three models, we observe no loss in explanatory power with the removal of D and the interaction term. Selecting only the best fit model ($assimilation \sim pressure * D$), we can see that a large majority of the variance (~94.4%) is explained only by pressure. The other two effects (D and the interaction between *pressure* and D) don't have a significant impact on the model.



2.4.2 ANOVAs for Photosynthesis

Above, we showed that the interaction between D and *pressure* and the fixed effect of D don't increase the explanatory power of the response of assimilation. For the remaining parameters, namely transpiration, Ci:Ca, and stomatal conductance, we find that the best fit model is $response \sim pressure * D$. The results of the anovas for those variables are found in Section 3.6.1.

What we observe there is that the interaction between D and *pressure* is fundamental in being able to explain our results for the remaining responses. Importantly (seen in the ANOVAs for the individual best-fit model) we see that D has no impact by itself, rather it modulates the impact of *pressure*. We can't aggregate for pressure in these cases due to the importance of the interaction term, rather we should try to understand how D and *pressure* interact, looking at the slope of their interaction.

2.4.3 Pressure : D interaction

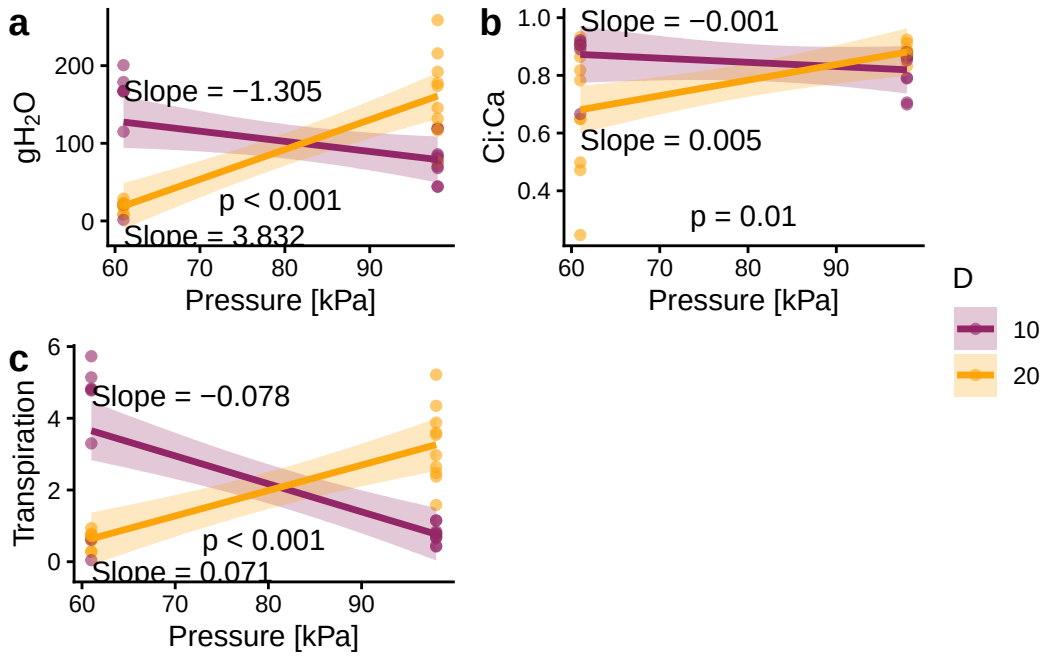


Figure 3: Responses of **a** stomatal conductance [mol m^{-2} s^{-1}], **b** intercellular to ambient CO_2 concentration, and **c** transpiration [mol H_2O m^{-2} s^{-1}] to pressure under different D. Shaded areas are the 95% confidence interval of the best-fit line; slopes come from the respective models (see above).

While the response changes with pressure, the direction of that change depends on D. We observe that pressure increases gas exchange, transpiration, and Ci:Ca under high D (low air water content) and decreases the responses of these variables under low D (high air water content).

Since it doesn't make sense to aggregate for pressure and the main results are presented in the above graphs, the treatment level results are presented in supplementary materials (Section 3.6.2).

2.5 Pigments (chlorophyll data)

2.5.1 Chlorophyll A:B ratio

There is no statistical evidence of an interaction between *pressure* and D (see Section 3.7.1), so we can drop all models which include *pressure* * D.

Looking at the chlorophyll A:B ratio:

```
# Analysis of Variance Table
#
# Model 1: chlab ~ pressure + time
# Model 2: chlab ~ pressure + D + time
#   Res.Df     RSS Df Sum of Sq      F Pr(>F)
# 1     37 0.22637
# 2     36 0.22398  1  0.002387 0.3837 0.5396
# Analysis of Variance Table
#
# Model 1: chlab ~ pressure + time
# Model 2: chlab ~ pressure * time
#   Res.Df     RSS Df Sum of Sq      F  Pr(>F)
# 1     37 0.22637
# 2     36 0.20635  1  0.020026 3.4938 0.06975 .
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# Analysis of Variance Table
#
# Response: chlab
#               Df   Sum Sq  Mean Sq F value   Pr(>F)
# pressure       1 0.061231 0.061231 10.6825 0.002383 **
# time           1 0.002031 0.002031  0.3543 0.555428
# pressure:time  1 0.020026 0.020026  3.4938 0.069753 .
# Residuals     36 0.206346 0.005732
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The chlorophyll A:B ratio seems to be influenced mostly by pressure with a slight, but non significant effect of the interaction of *time*. This seems to indicate that for this variable we could aggregate pressure treatments and time, but returning to our "pseudoreplication" issue, we will maintain our two sampling times separate.

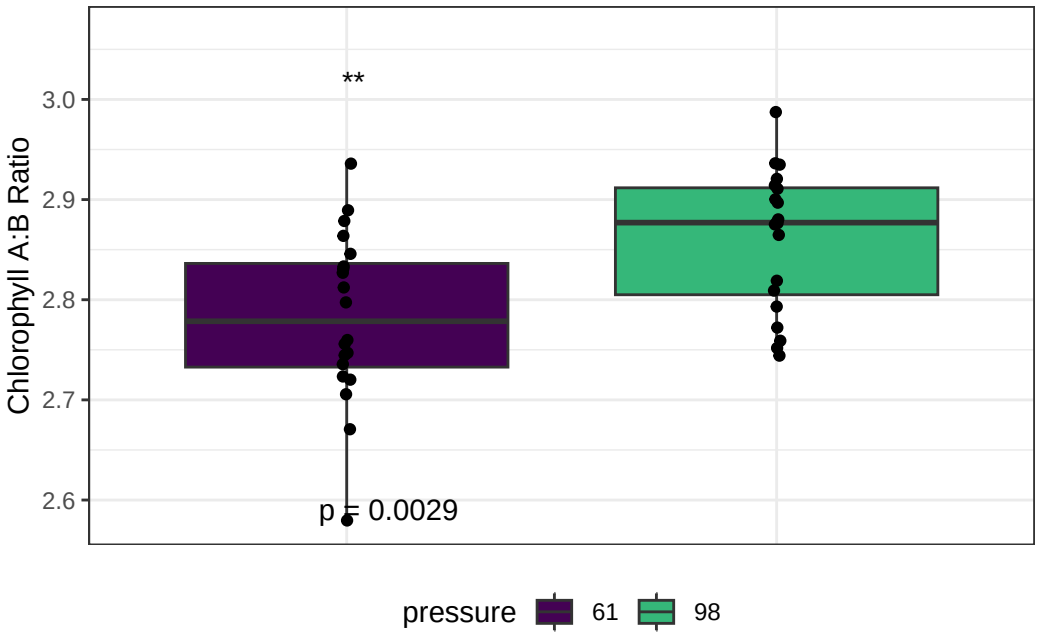


Figure 4: Chlorophyll A:B ratio, aggregated by pressure treatment.

A:B is usually used as an indicator of light availability, with shade-acclimated leaves often showing a lower ratio due to a larger investment in light-harvesting complexes. It can also be interpreted as an attempt to maximize light capture. In our scenario, this seems to indicate that at lower pressure, *Arabidopsis* plants are investing more in light harvesting than processing machinery - a potential response to the increased efficency of the photosynthetic apparatus outlined in Section 2.3.

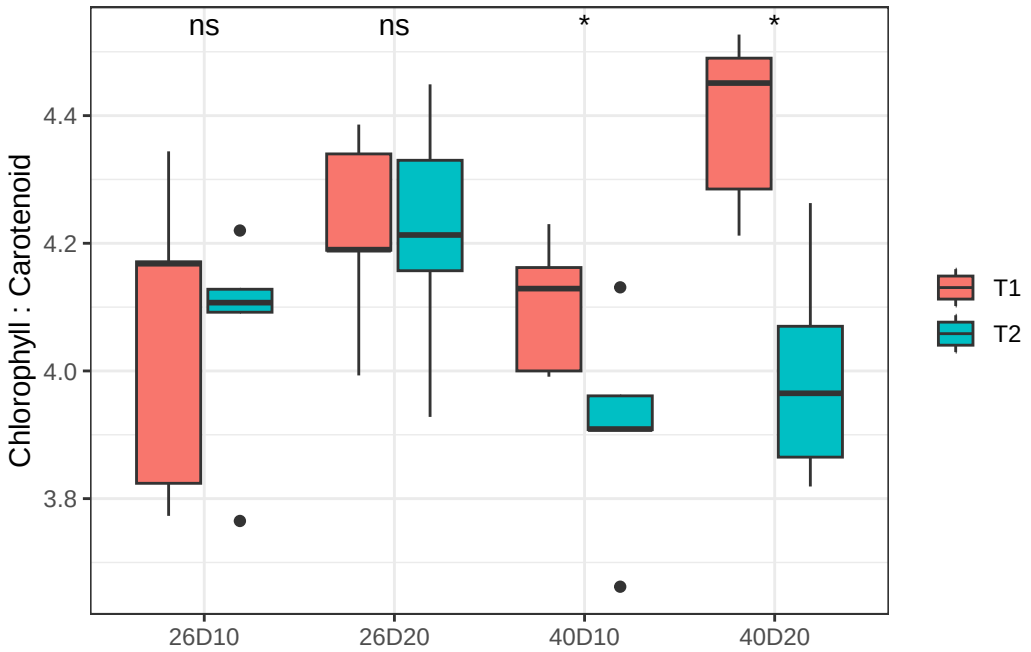
2.5.2 Chlorophyll:carotenoid ratio & chlorophyll density

Looking more closely at the chloropyhll:carotenoid ratio:

```
# Analysis of Variance Table
#
# Model 1: chlcar ~ D + time
# Model 2: chlcar ~ pressure + D + time
#   Res.Df    RSS Df Sum of Sq      F Pr(>F)
# 1     37 1.2551
# 2     36 1.2416  1  0.013579 0.3937 0.5343
# Analysis of Variance Table
#
# Model 1: chlcar ~ D + time
# Model 2: chlcar ~ D * time
#   Res.Df    RSS Df Sum of Sq      F Pr(>F)
# 1     37 1.2551
```

```
# 2     36 1.2252  1  0.029976 0.8808 0.3542
# Analysis of Variance Table
#
# Response: chlcar
#           Df  Sum Sq  Mean Sq F value   Pr(>F)
# D          1 0.29705 0.297045  8.7565 0.005354 **
# time       1 0.21214 0.212139  6.2536 0.016951 *
# Residuals 37 1.25515 0.033923
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We have a different effect - *pressure* no longer seems to play a role, and rather we have a strong effect of D and a minimal effect of *time*, though their interaction does not have a statistically significant impact. It could be interesting to delve into this temporal trend. To do so, let's look at a temporal comparison based on treatments.

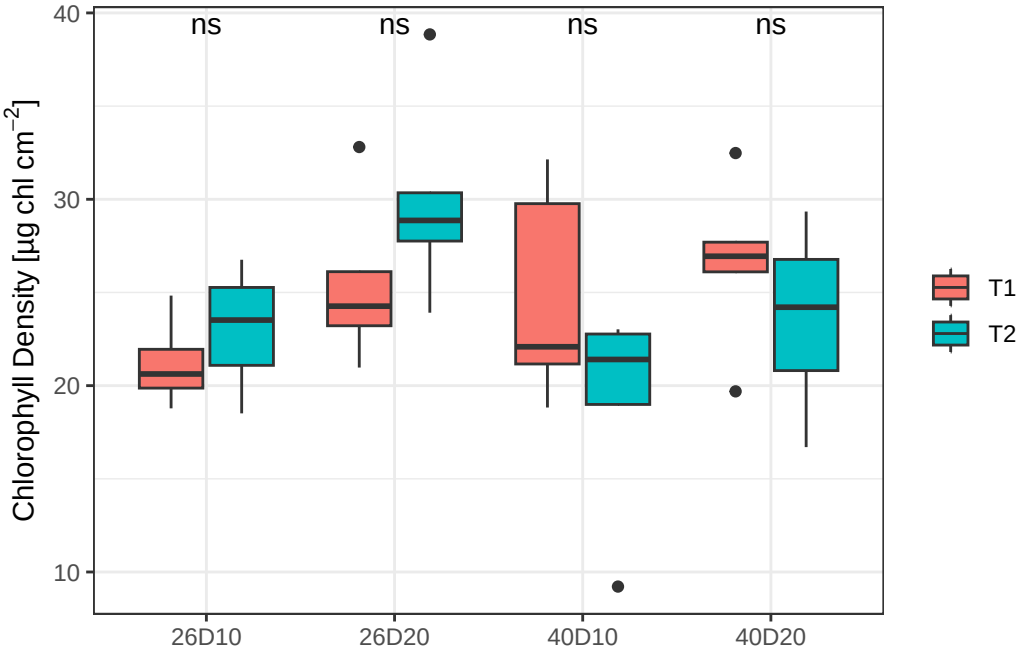


Cool! There's a statistically significant decrease in the ratio at low pressure with time. This could indicate a decrease in stress response as carotenoids are photoprotective pigments and can indicate an investment in stress tolerance. To confirm this interpretation, we have to look at the chlorophyll density trends:

```
# Analysis of Variance Table
#
# Model 1: density ~ D + time
# Model 2: density ~ pressure + D + time
#   Res.Df    RSS Df Sum of Sq      F Pr(>F)
# 1     37 910.87
```

```
# 2     36 891.04  1    19.833 0.8013 0.3766
# Analysis of Variance Table
#
# Model 1: density ~ D + time
# Model 2: density ~ D * time
#   Res.Df    RSS Df Sum of Sq      F Pr(>F)
# 1     37 910.87
# 2     36 892.96  1    17.911 0.7221 0.4011
# Analysis of Variance Table
#
# Response: density
#           Df Sum Sq Mean Sq F value   Pr(>F)
# D          1 190.41 190.407  7.7344 0.008472 **
# time       1   3.73   3.726  0.1514 0.699475
# Residuals 37 910.87  24.618
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The case for chlorophyll density is similar, but cleaner, with the only fixed effect having a significant impact on the variable being *D*. Time doesn't seem to have an impact. But let's double check, to understand our chlorophyll:carotenoid ratios better.



While we seem to observe a temporal trend - with an increase of chlorophyll at high pressure and a decrease at low pressure over time, it is not statistically significant. This lack of statistical significance is further supported by the anovas above, indicating that the most likely interpretation of the temporal signal in the chlorophyll:carotenoid ratio is a decrease in carotenoids. Thus a decreased stress response - since the decrease in the

chlorophyll:carotenoid ratio is only observed in the low pressure treatments, this is potentially an adaptation to pressure.

The original treatment-level plots can be found Section 3.7.

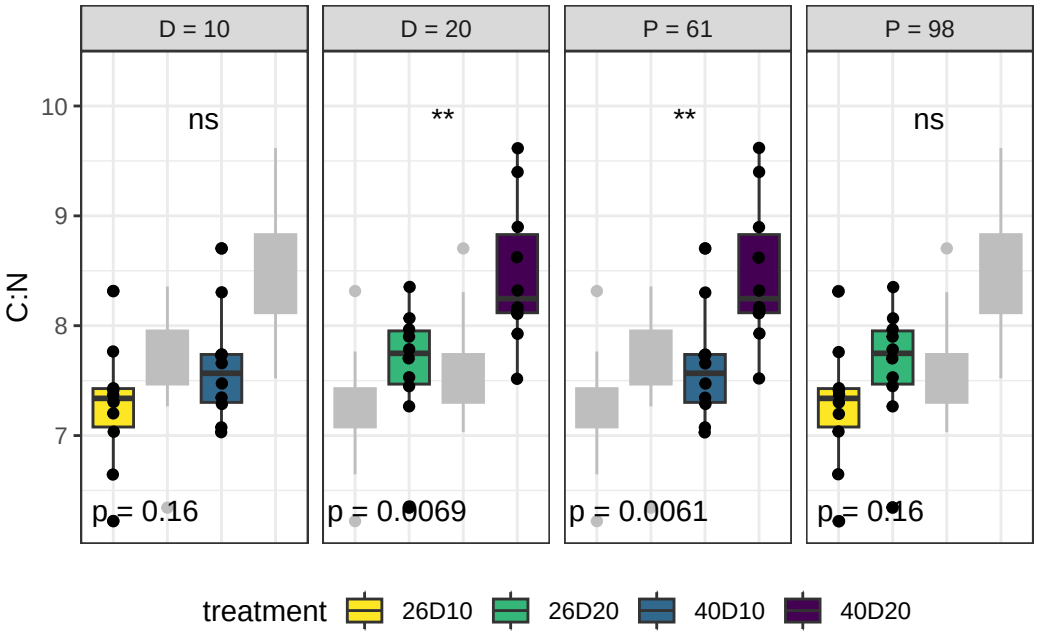
2.6 Carbon content - C:N ratio and d13C

2.6.1 C:N Ratio

```
# Analysis of Variance Table
#
# Model 1: cn ~ pressure
# Model 2: cn ~ pressure + D
# Model 3: cn ~ pressure + D + pressure * D
#   Res.Df    RSS Df Sum of Sq       F    Pr(>F)
# 1     58 34.941
# 2     57 26.509  1    8.4312 18.5085 6.847e-05 ***
# 3     56 25.510  1    0.9995  2.1941    0.1441
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# Analysis of Variance Table
#
# Model 1: cn ~ D
# Model 2: cn ~ pressure + D
# Model 3: cn ~ pressure + D + pressure * D
#   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
# 1     58 29.938
# 2     57 26.509  1    3.4282 7.5258 0.008155 **
# 3     56 25.510  1    0.9995 2.1941 0.144147
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# Analysis of Variance Table
#
# Response: cn
#           Df  Sum Sq Mean Sq F value    Pr(>F)
# pressure   1  3.4282  3.4282  7.3714  0.008754 **
# D          1  8.4312  8.4312 18.1287 7.799e-05 ***
# Residuals 57 26.5093  0.4651
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There is no statistical evidence of an interaction between *pressure* and D, but in both anova model comparisons, the model including both *pressure* and D performs much better than the one with a single effect. The anova of the single selected model - $cn \sim pressure + D$,

indicates that D has a stronger effect than pressure (F = 18, F = 7). This means we cannot aggregate for pressure treatment (or for D treatment). The original graphs follow:



There seems to be a modulating signal of *D* on the effect of *pressure* on cn (also observed below in the d13C). Generally, we observe an increase in C:N with a decrease in pressure an increase in D. This signal is significant only in the more stressed plants (i.e. low pressure - P = 61, and high D - D = 20), but is present across all analyses.

2.6.2 $\delta^{13}C$

```
# Analysis of Variance Table
#
# Model 1: C_discr ~ pressure
# Model 2: C_discr ~ pressure + D
# Model 3: C_discr ~ pressure + D + pressure * D
#   Res.Df     RSS Df Sum of Sq      F    Pr(>F)
# 1     38 18.0135
# 2     37  8.0156  1    9.9979 80.380 1.056e-10 ***
# 3     36  4.4778  1    3.5378 28.443 5.410e-06 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# Analysis of Variance Table
#
# Model 1: C_discr ~ D
# Model 2: C_discr ~ pressure + D
# Model 3: C_discr ~ pressure + D + pressure * D
```

```
#   Res.Df     RSS Df Sum of Sq       F    Pr(>F)
# 1     38 27.5423
# 2     37  8.0156  1   19.5267 156.988 1.089e-14 ***
# 3     36  4.4778  1    3.5378  28.443 5.410e-06 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# Analysis of Variance Table
#
# Response: C_discr
#             Df  Sum Sq Mean Sq F value    Pr(>F)
# pressure     1 19.5267 19.5267 156.988 1.089e-14 ***
# D            1  9.9979  9.9979  80.380 1.056e-10 ***
# pressure:D   1  3.5378  3.5378  28.443 5.410e-06 ***
# Residuals   36  4.4778  0.1244
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Pressure, D, and their interaction have strong effects on $\delta^{13}C$. We sadly then cannot aggregate by either *pressure* or D, but can take a look at the interaction term as well as a closer look at how each effect impacted $\delta^{13}C$.

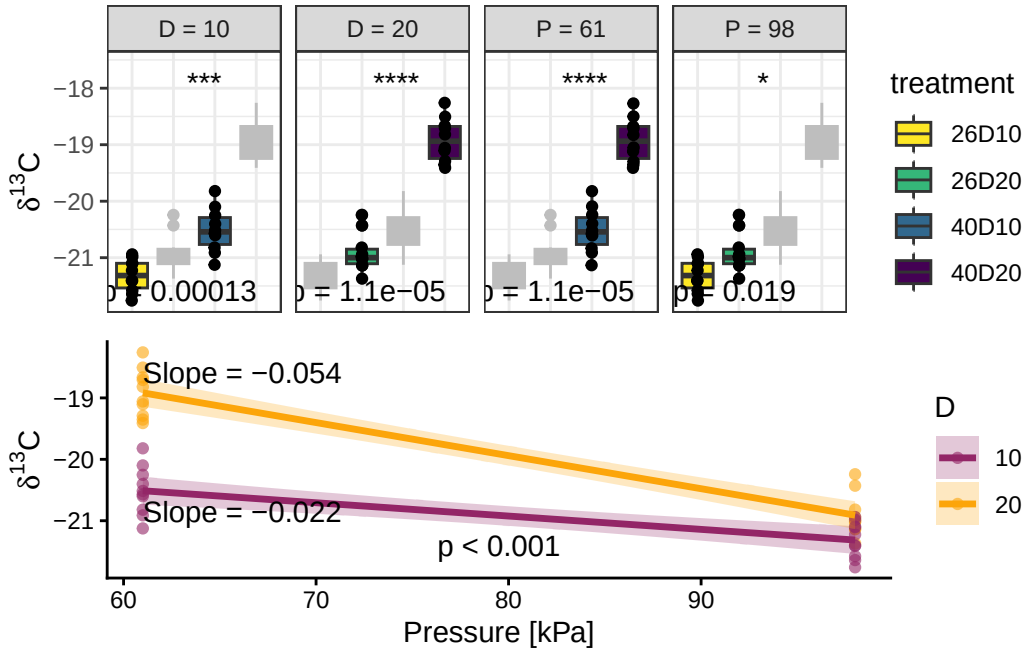


Figure 5: ^{13}C discrimination normalized for $\delta^{13}C$ of the air in each chamber.

D has a significant modulating effect on the impact of pressure ($p < 0.001$), with higher D (drier conditions) increasing the effect of *pressure* on the discrimination of ^{13}C. $\delta^{13}C$ is understood as a long term physiologic signal for intrinsic water use efficiency, in this

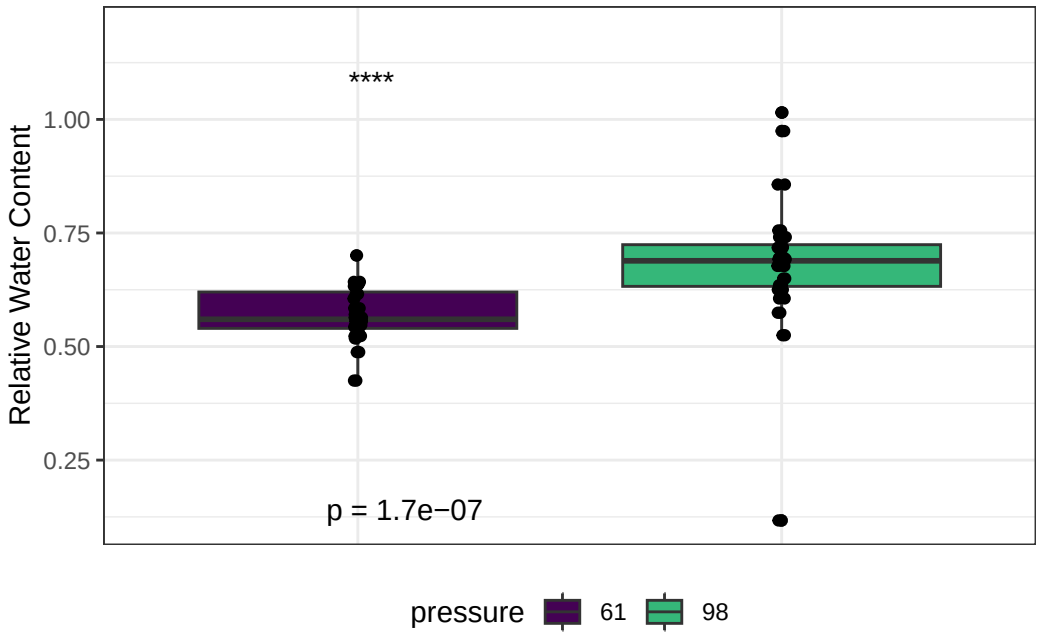
case helping us develop an understanding of the long term stomatal conductance and photosynthetic functioning. Plants at low pressure discriminate less against $13C$, indicating a potential signal of long-term decreased stomatal conductance. Under dry conditions, this effect is amplified. Our C:N results - with a higher C:N in our low pressure plants, supports this stomatal-conductance based interpretation.

2.7 Biometric Data

2.7.1 Relative Water Content

```
# Analysis of Variance Table
#
# Model 1: RWC ~ pressure
# Model 2: RWC ~ pressure + D
# Model 3: RWC ~ pressure + D + pressure * D
#   Res.Df     RSS Df  Sum of Sq      F Pr(>F)
# 1     38 0.68221
# 2     37 0.68196  1 0.00025124 0.0133 0.9089
# 3     36 0.68093  1 0.00103083 0.0545 0.8167
# Analysis of Variance Table
#
# Model 1: RWC ~ D
# Model 2: RWC ~ pressure + D
# Model 3: RWC ~ pressure + D + pressure * D
#   Res.Df     RSS Df Sum of Sq      F  Pr(>F)
# 1     38 0.80607
# 2     37 0.68196  1  0.124116 6.5619 0.01475 *
# 3     36 0.68093  1  0.001031 0.0545 0.81673
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# Analysis of Variance Table
#
# Response: RWC
#           Df  Sum Sq  Mean Sq F value  Pr(>F)
# pressure   1 0.12412 0.124116  6.9134 0.01228 *
# Residuals 38 0.68221 0.017953
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The *pressure* signal for relative water content is very strong, indicating that we can aggregate for pressure treatments.



At lower pressure we have a significantly lower RWC, giving us long-term image of the impact of our treatments. This is biometric support of our interpretation thus far that at low pressure *Arabidopsis thaliana* reduces stomatal conductance.

There are no notable trends in the other biometric results, but they are presented in Section 3.8 for your parousal.

3 Supplementary Graphics

Here we include some supplementary graphics, tables, and analyses for context.

3.1 ANOVA for Chlorophyll Fluorescence

Above, Section 2.3.1, we showed that the interaction between D and *pressure* and the fixed effect of D don't increase the explanatory power of models explaining the response of chlorophyll fluorescence to our treatments using the case study of *PSII (Y)*. Below we show the ANOVA of the remaining responses tested. We see the same trend, where D, *pressure*, their interaction, and often even *time* don't increase the explanatory value of the models.

```
# [1] "ANOVA for NPQ1"
# Analysis of Variance Table
#
# Model 1: get(response) ~ irr + pressure + D + pressure * D + time
# Model 2: get(response) ~ irr + pressure + D + time
# Model 3: get(response) ~ irr + pressure + time
# Model 4: get(response) ~ irr + pressure
#   Res.Df    RSS Df Sum of Sq      F Pr(>F)
# 1    594 64.216
# 2    595 64.249 -1  -0.03289 0.3043 0.5814
# 3    596 64.335 -1  -0.08613 0.7967 0.3725
# 4    597 64.711 -1  -0.37566 3.4749 0.0628 .
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# [1] "ANOVA for ETR"
# Analysis of Variance Table
#
# Model 1: get(response) ~ irr + pressure + D + pressure * D + time
# Model 2: get(response) ~ irr + pressure + D + time
# Model 3: get(response) ~ irr + pressure + time
# Model 4: get(response) ~ irr + pressure
#   Res.Df    RSS Df Sum of Sq      F Pr(>F)
# 1    594 355710
# 2    595 355763 -1    -53.81 0.0899 0.7645
# 3    596 356210 -1   -446.66 0.7459 0.3881
# 4    597 357283 -1  -1073.22 1.7922 0.1812
# [1] "ANOVA for qE"
# Analysis of Variance Table
#
# Model 1: get(response) ~ irr + pressure + D + pressure * D + time
# Model 2: get(response) ~ irr + pressure + D + time
# Model 3: get(response) ~ irr + pressure + time
```

```
# Model 4: get(response) ~ irr + pressure
#   Res.Df    RSS Df Sum of Sq      F Pr(>F)
# 1    594 56.574
# 2    595 56.636 -1 -0.061352 0.6442 0.4225
# 3    596 56.679 -1 -0.043300 0.4546 0.5004
# 4    597 56.765 -1 -0.085951 0.9024 0.3425
# [1] "ANOVA for qP"
# Analysis of Variance Table
#
# Model 1: get(response) ~ irr + pressure + D + pressure * D + time
# Model 2: get(response) ~ irr + pressure + D + time
# Model 3: get(response) ~ irr + pressure + time
# Model 4: get(response) ~ irr + pressure
#   Res.Df    RSS Df  Sum of Sq      F Pr(>F)
# 1    594 39.096
# 2    595 39.101 -1 -0.0057562 0.0875 0.7675
# 3    596 39.108 -1 -0.0067901 0.1032 0.7482
# 4    597 39.116 -1 -0.0081588 0.1240 0.7249
```

3.2 P-values for Chlorophyll Fluorescence graphs

Table 1: Pairwise p-values for PSII chlorophyll fluorescence.

comp1	comp2	irr	p.value	sig	n1	n2	stat	time
61	98	150	0.0000010	***	20	20	t	T1
61	98	300	0.0000072	***	20	20	w	T1
61	98	600	0.0036451	**	20	20	t	T1
61	98	1200	0.0001537	***	20	20	w	T1
61	98	2000	0.0082554	**	20	20	t	T1
61	98	0	0.5065868		20	20	w	T1
61	98	150	0.3543989		20	30	w	T2
61	98	300	0.0000046	***	20	30	t	T2
61	98	600	0.0051697	**	20	30	t	T2
61	98	1200	0.0016923	**	20	30	t	T2
61	98	2000	0.5330057		20	30	t	T2
61	98	0	0.0000000	***	20	30	w	T2

Table 2: Pairwise p-values for non-photochemical quenching (NPQ).

comp1	comp2	irr	p.value	sig	n1	n2	stat	time
61	98	150	0.1287705		20	20	t	T1
61	98	300	0.0001765	***	20	20	w	T1

61	98	600	0.0000336	***	20	20	w	T1
61	98	1200	0.0000001	***	20	20	t	T1
61	98	2000	0.0365463	*	20	20	t	T1
61	98	0	1.0000000		20	20	none	T1
61	98	150	0.7166095		20	30	w	T2
61	98	300	0.0000010	***	20	30	t	T2
61	98	600	0.0054492	**	20	30	t	T2
61	98	1200	0.0042824	**	20	30	t	T2
61	98	2000	0.0001755	***	20	30	t	T2
61	98	0	1.0000000		20	30	none	T2

Table 3: Pairwise p-values for electron transport rate (ETR).

comp1	comp2	irr	p.value	sig	n1	n2	stat	time
61	98	150	0.0000010	***	20	20	t	T1
61	98	300	0.0000072	***	20	20	w	T1
61	98	600	0.0036451	**	20	20	t	T1
61	98	1200	0.0001537	***	20	20	w	T1
61	98	2000	0.0082554	**	20	20	t	T1
61	98	0	1.0000000		20	20	none	T1
61	98	150	0.3543989		20	30	w	T2
61	98	300	0.0000046	***	20	30	t	T2
61	98	600	0.0051697	**	20	30	t	T2
61	98	1200	0.0016923	**	20	30	t	T2
61	98	2000	0.5330057		20	30	t	T2
61	98	0	1.0000000		20	30	none	T2

Table 4: Pairwise p-values for energy-dependent quenching (qE).

comp1	comp2	irr	p.value	sig	n1	n2	stat	time
61	98	150	0.0002006	***	20	20	w	T1
61	98	300	0.0001551	***	20	20	w	T1
61	98	600	0.0011601	**	20	20	w	T1
61	98	1200	0.0021836	**	20	20	t	T1
61	98	2000	0.1928824		20	20	t	T1
61	98	0	1.0000000		20	20	none	T1
61	98	150	0.0179431	*	20	30	w	T2
61	98	300	0.0000002	***	20	30	t	T2
61	98	600	0.0114656	*	20	30	t	T2
61	98	1200	0.0000613	***	20	30	w	T2

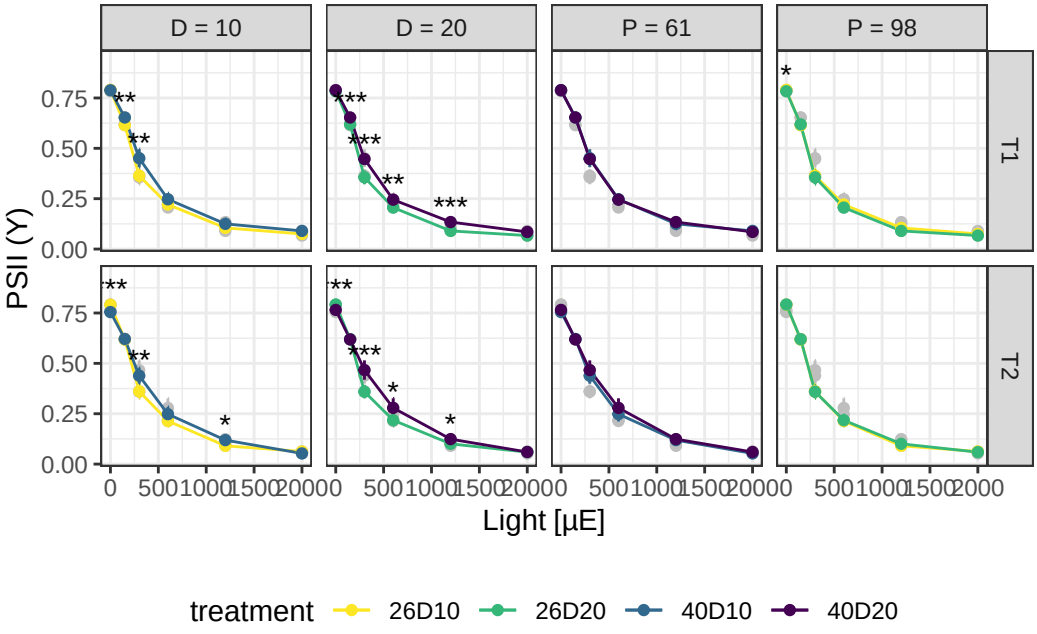
61	98	2000	0.0554734	20	30	w	T2
61	98	0	1.0000000	20	30	none	T2

Table 5: Pairwise p-values for photochemical quenching (qP).

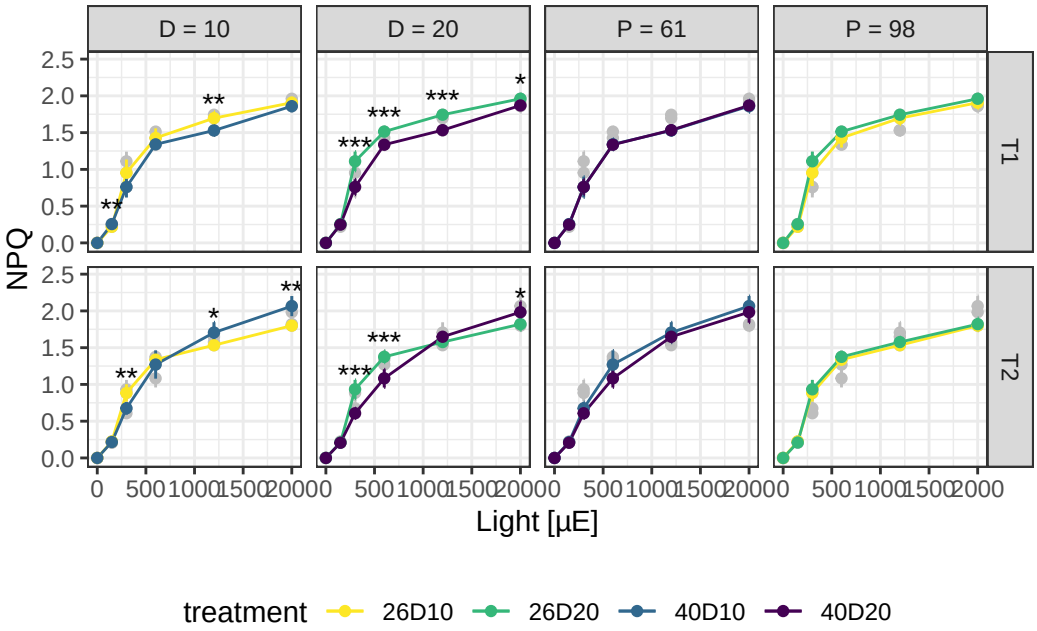
comp1	comp2	irr	p.value	sig	n1	n2	stat	time
61	98	150	0.0000002	***	20	20	w	T1
61	98	300	0.0000044	***	20	20	w	T1
61	98	600	0.0018865	**	20	20	t	T1
61	98	1200	0.0002540	***	20	20	w	T1
61	98	2000	0.0080964	**	20	20	t	T1
61	98	0	1.0000000		20	20	none	T1
61	98	150	0.0257578	*	20	30	t	T2
61	98	300	0.0000602	***	20	30	w	T2
61	98	600	0.0013082	**	20	30	t	T2
61	98	1200	0.0001059	***	20	30	t	T2
61	98	2000	0.7761340		20	30	t	T2
61	98	0	1.0000000		20	30	none	T2

3.3 Chlorophyll Fluorescence - Unique Treatments

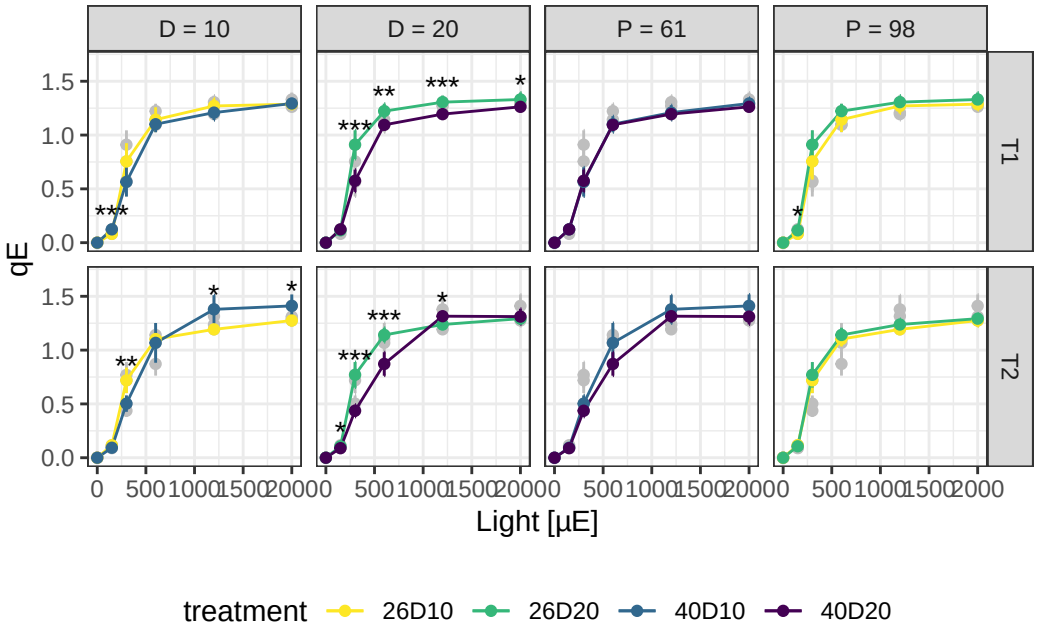
3.3.1 PSII(Y)



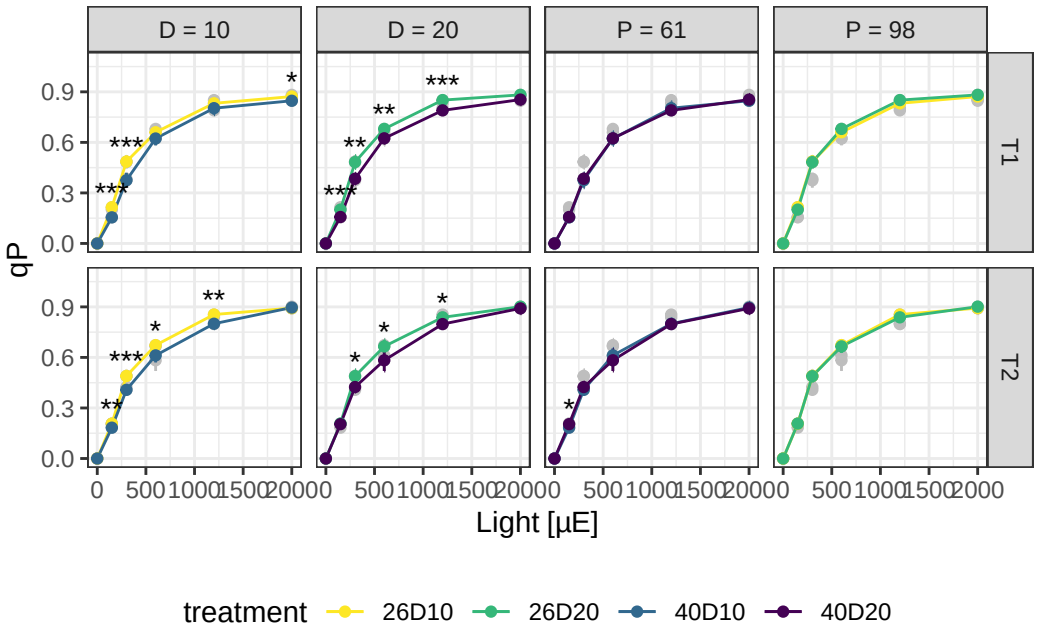
3.3.2 NPQ



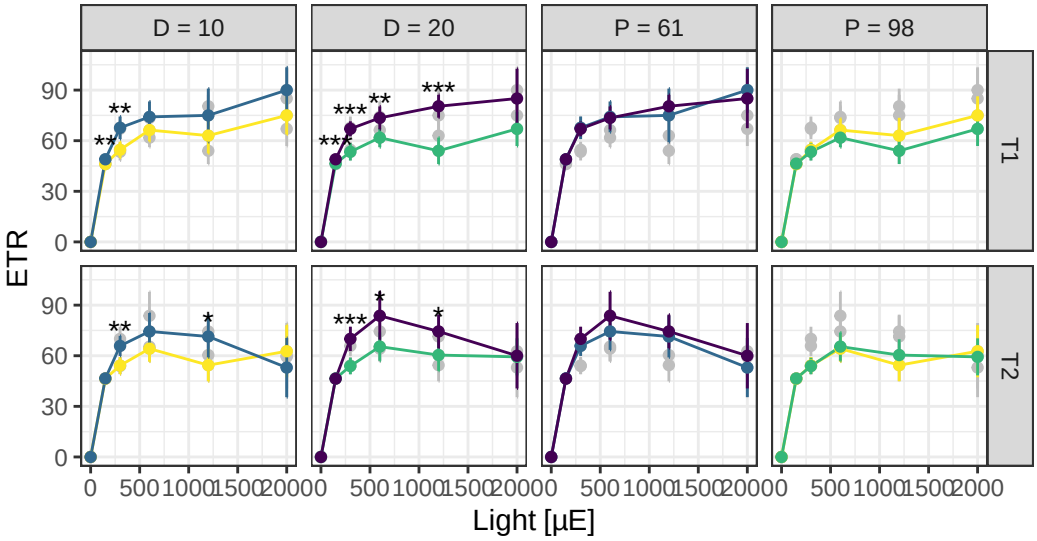
3.3.3 qE - rapid recovery of NPQ

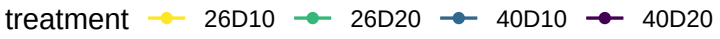


3.3.4 qP - open PSII reaction centers



3.3.5 ETR





3.4 Chlorophyll Fluorescence - Time-resolution data

3.4.1 PSII

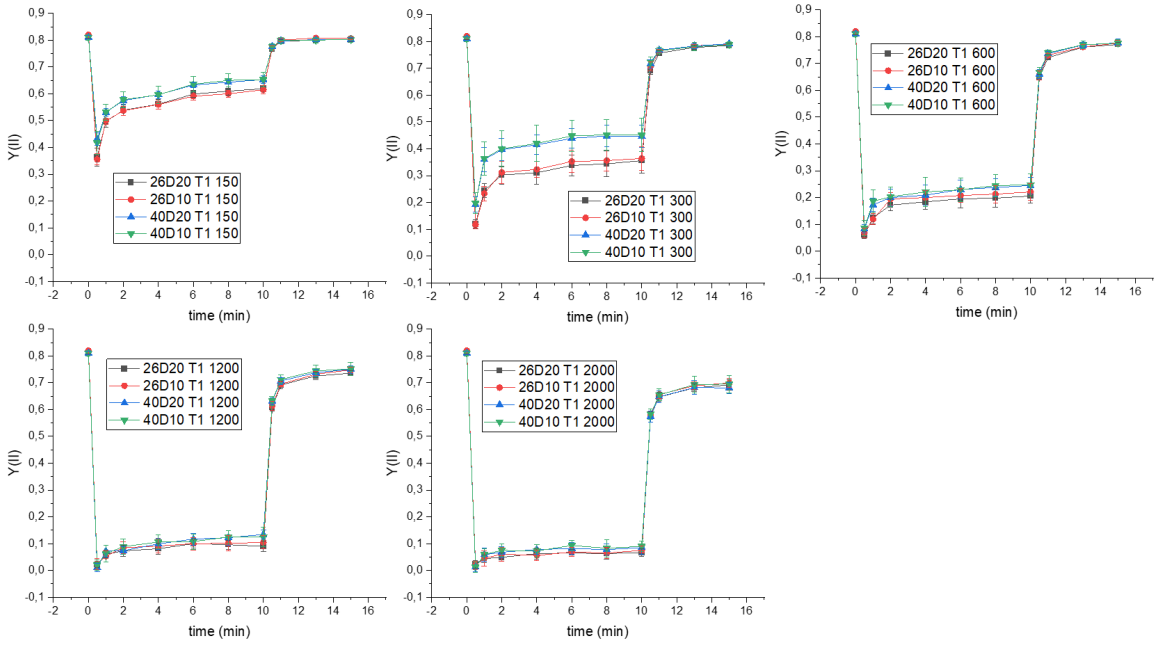


Figure 6: Quantum yield of PSII at a temporal scale across 5 light intensities, colors and symbols indicate treatments, light intensity [E] in each legend. Measurements taken halfway through the experiment- T1.

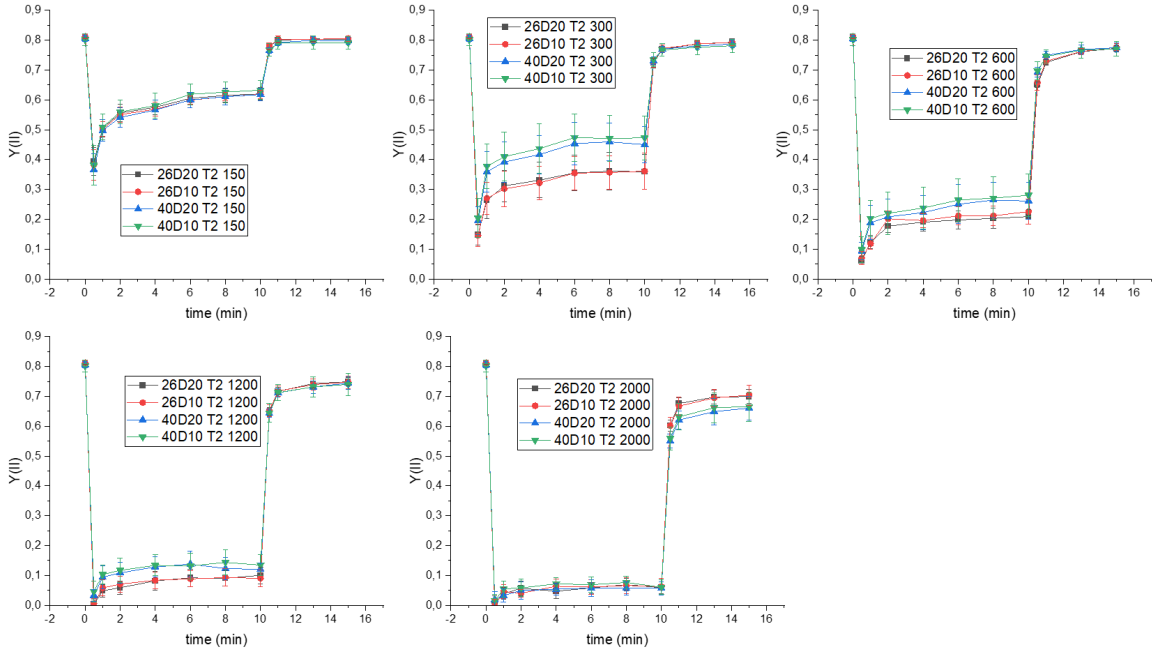


Figure 7: Quantum yield of PSII at T2. All other information same as Figure above.

3.4.2 NPQ

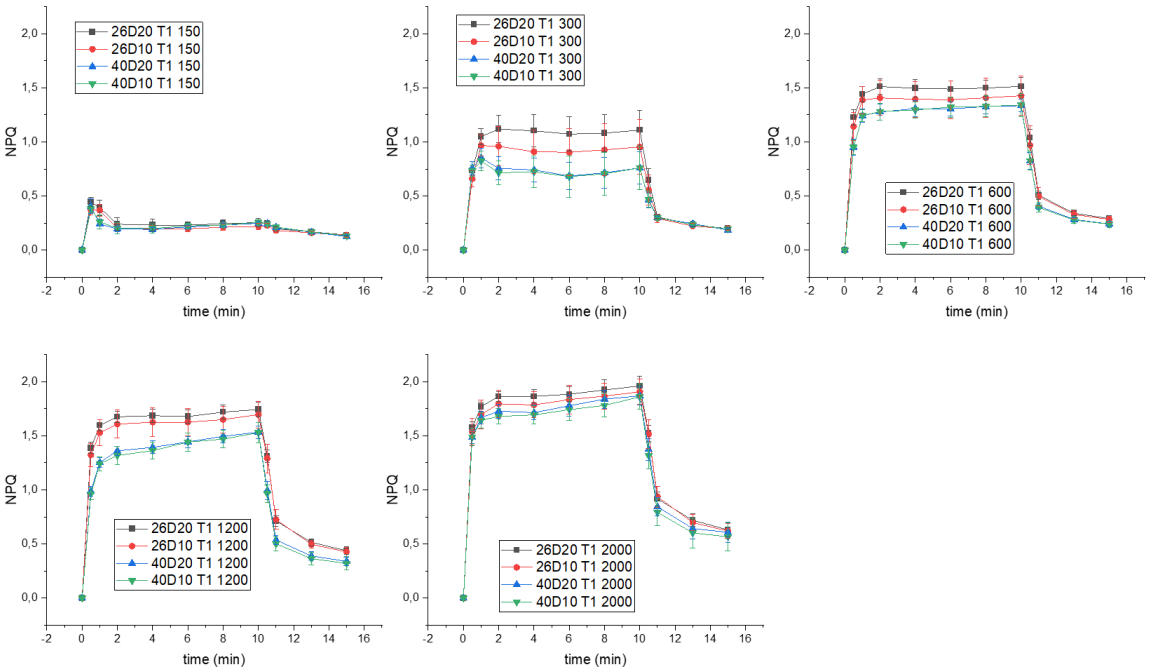


Figure 8: Non-photochemical quenching at T1 across different light intensities (see legend of each graph).

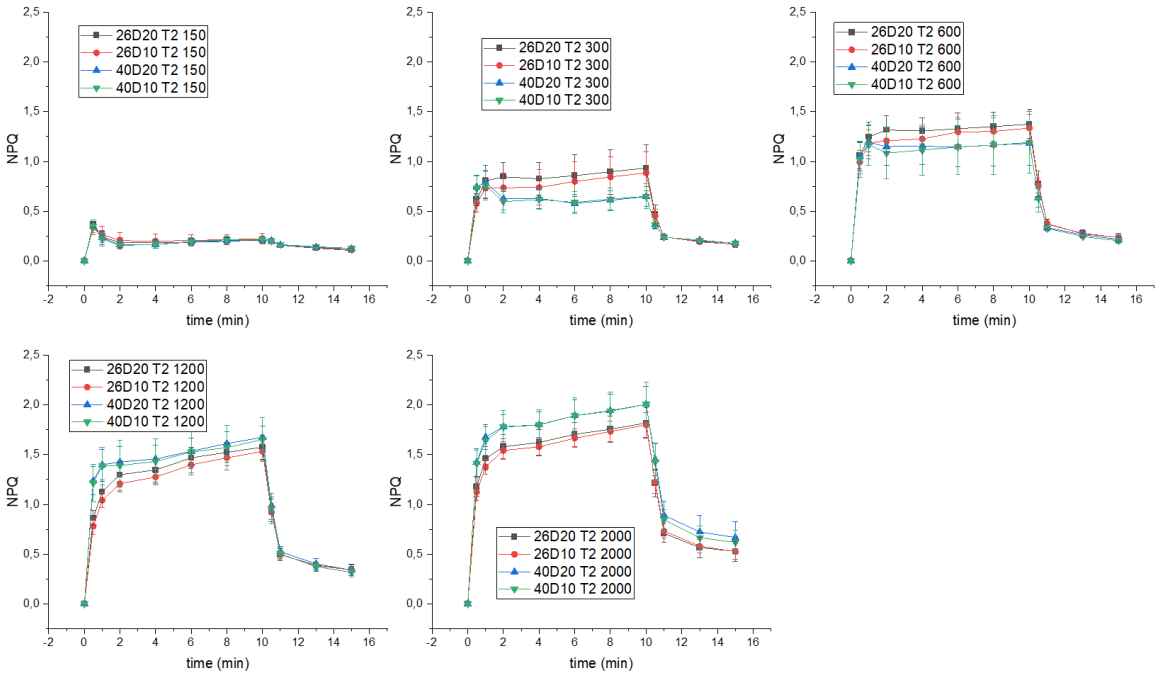


Figure 9: Non-photochemical quenching at T2.

3.4.3 ETR

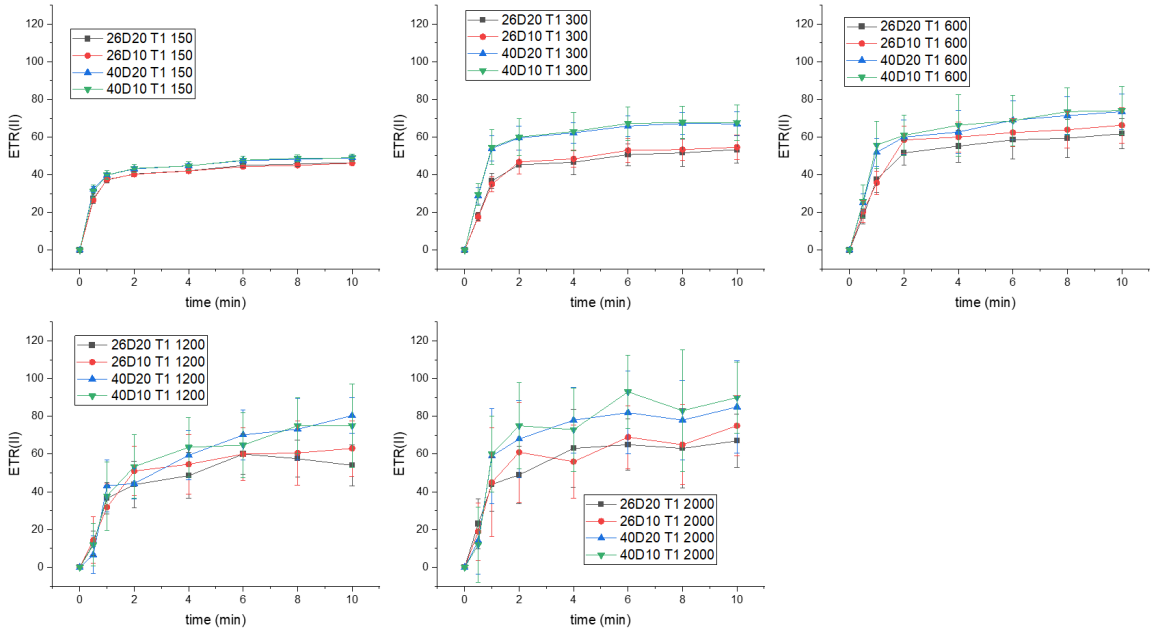


Figure 10: Electron transport rate at T1.

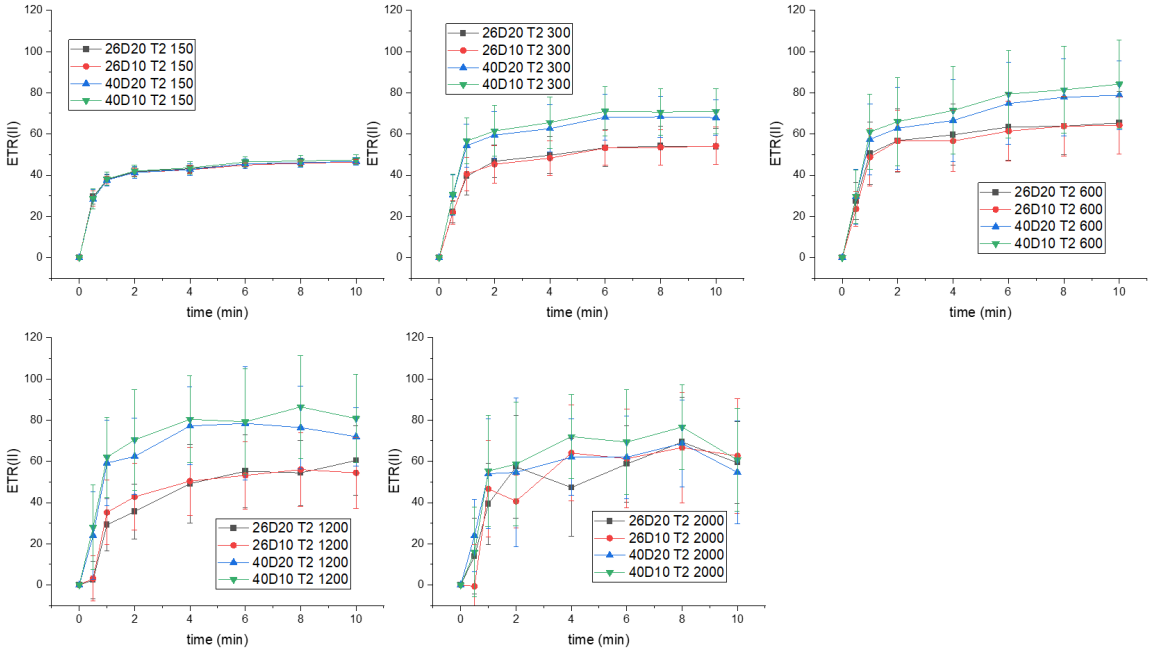


Figure 11: Electron transport rate at T2.

3.4.4 qP

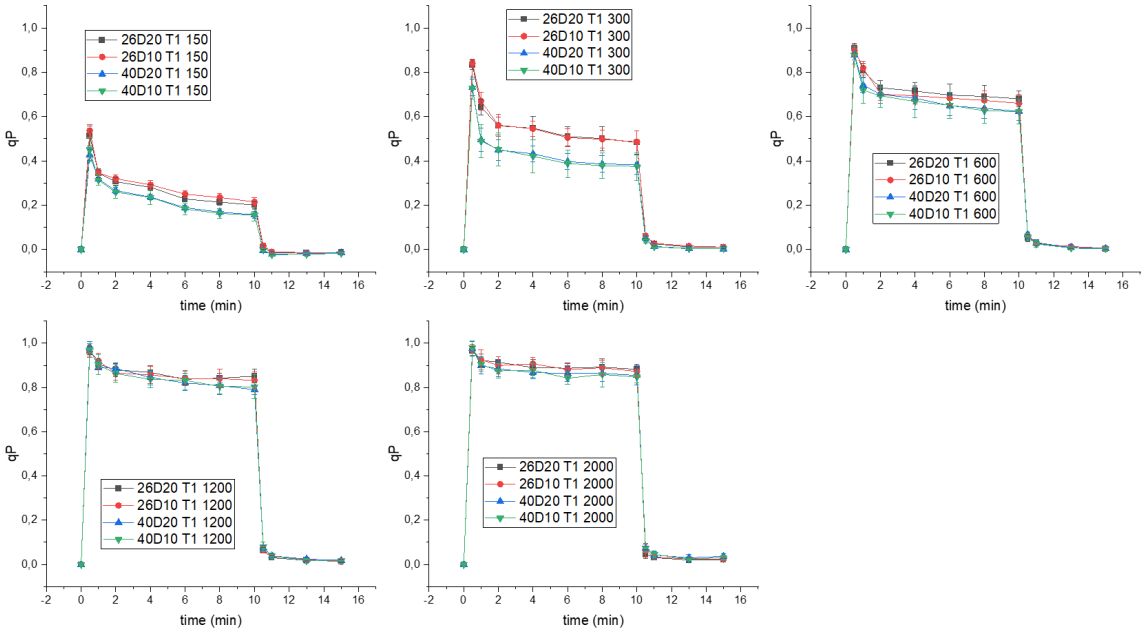


Figure 12: Photochemical quenching at T1.

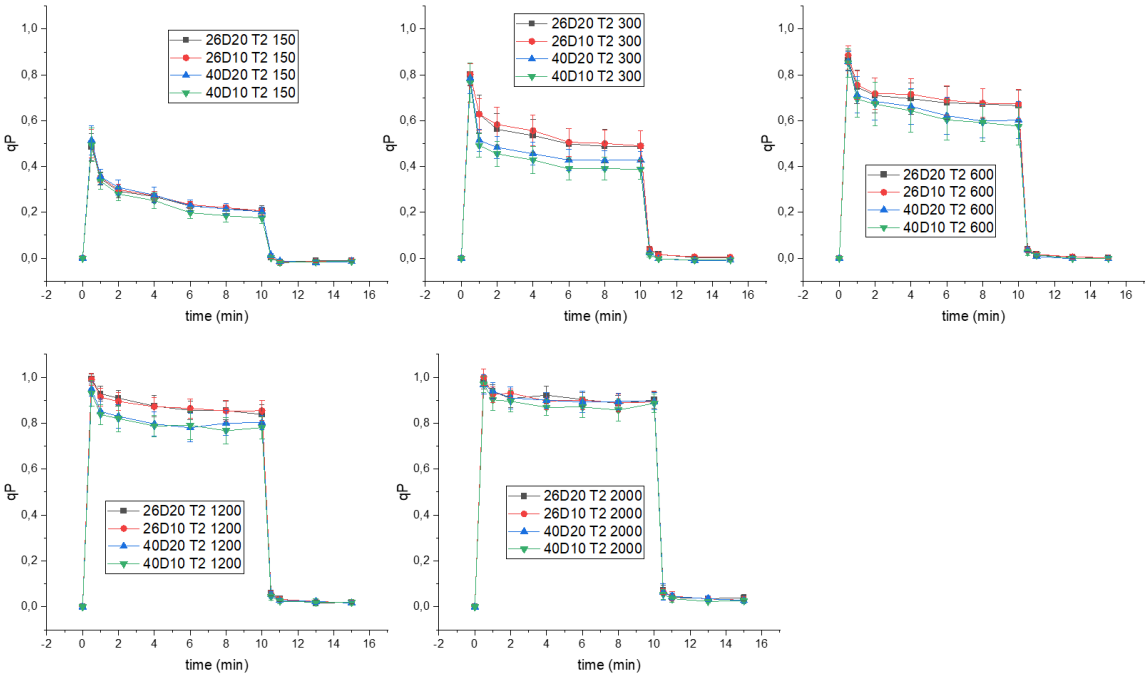
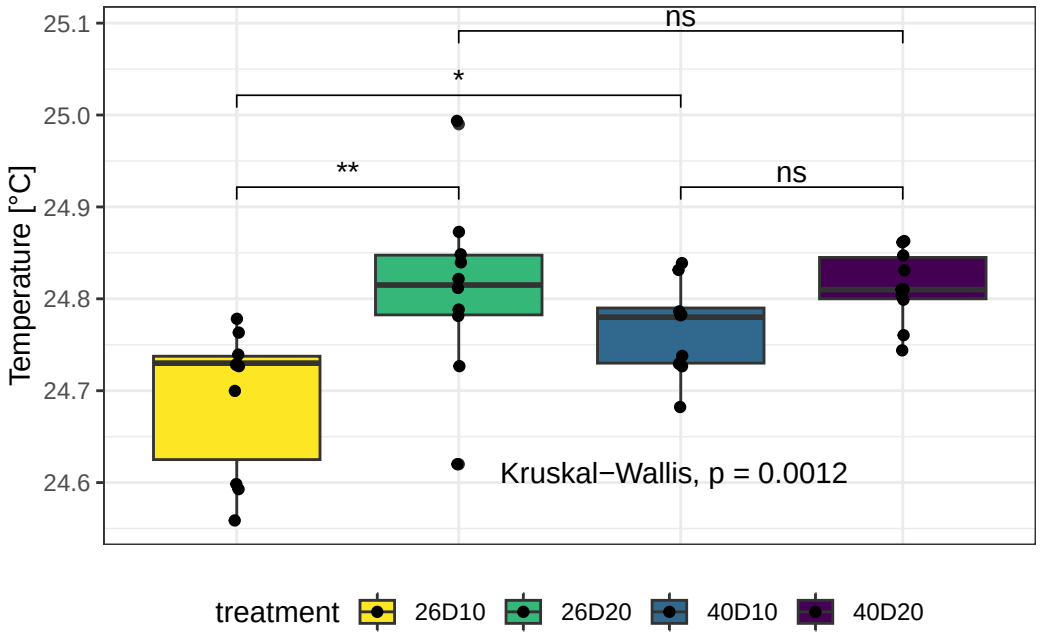


Figure 13: Photochemical quenching at T2.

Figures in this subsection (@sec-supp-chlotemp) courtesy of S. Cazzaniga.

3.5 Leaf temperature

The measurements shown in Section 2.4 are temperature dependent. Here we show the leaf temperature data during the measurements.


ns
*
**
ns
25.1
25.0
24.9
24.8
24.7
24.6
Temperature [°C]
Kruskal−Wallis, p = 0.0012
treatment
26D10
26D20
40D10
40D20

3.6 Photosynthesis

3.6.1 Anovas and Residual Plots for lms

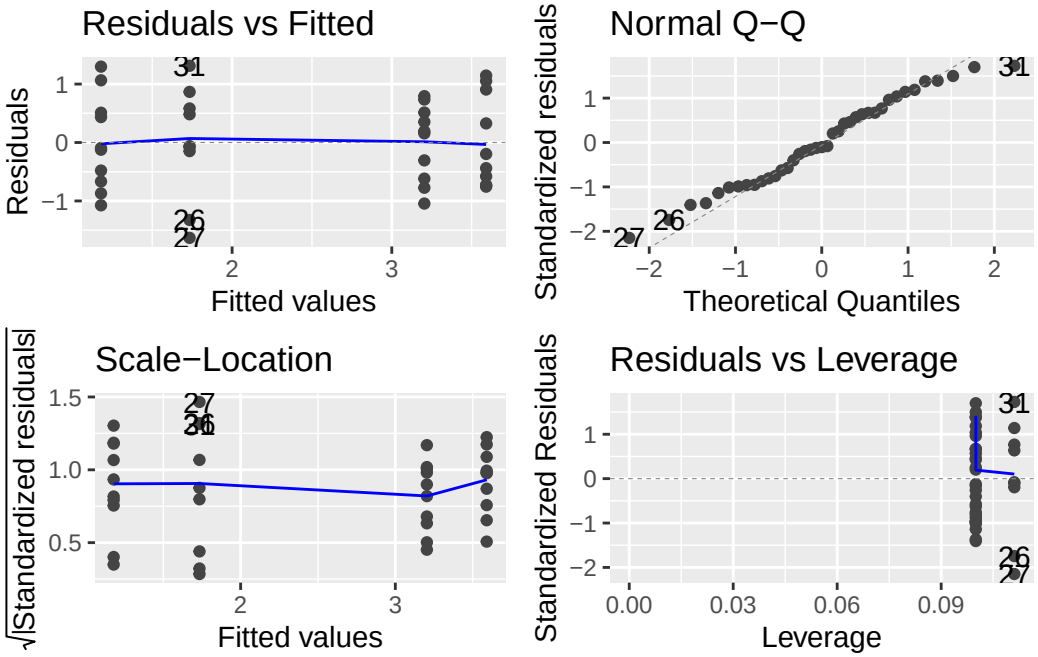
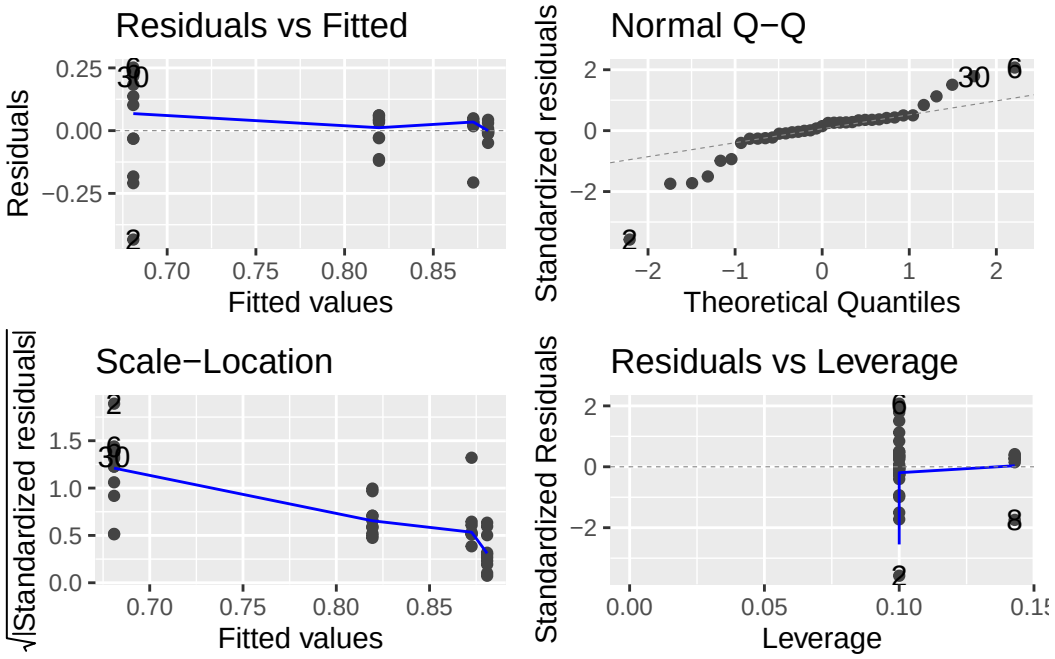
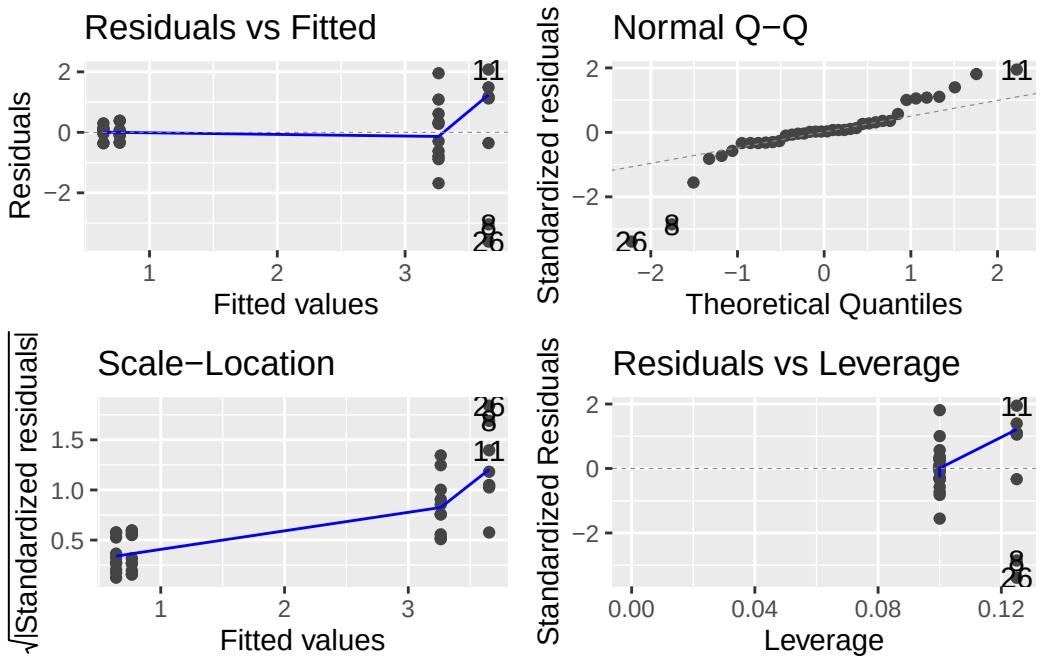


Figure 14: Autoplot for model: A ~ pressure * D. Normal Q-Q plots for remaining variables not shown, for space, but exhibit the same trend (consistent with a normal distribution of residuals, and thus supporting our use of linear models to model our data).

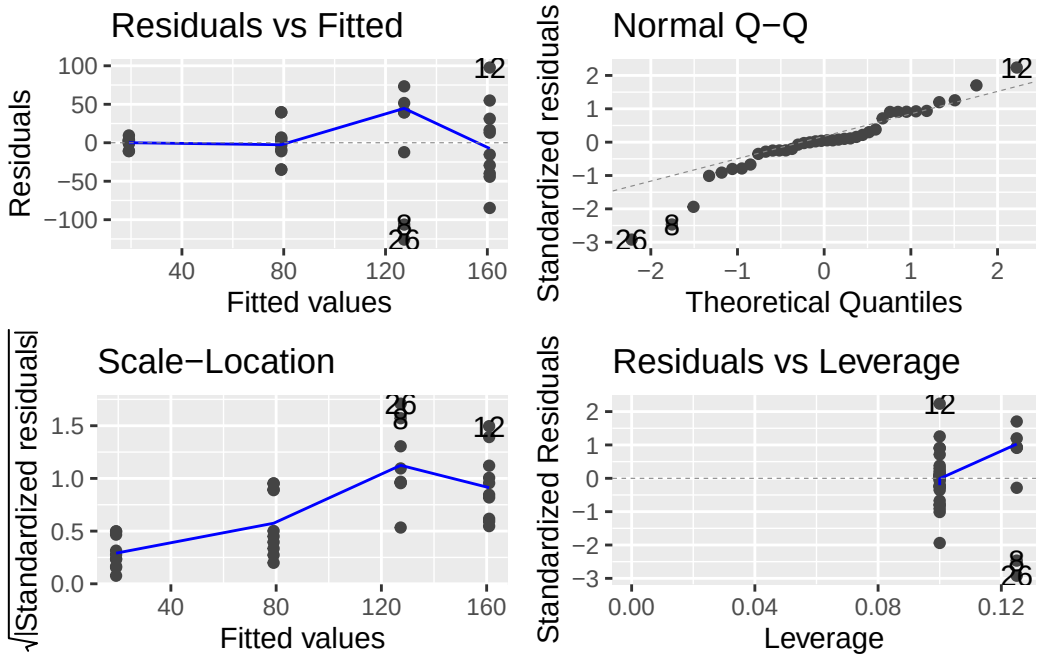
```
#
#  ANOVA for cica
#
# Analysis of Variance Table
#
# Model 1: get(response) ~ pressure
# Model 2: get(response) ~ pressure + D
# Model 3: get(response) ~ pressure + D + pressure * D
#   Res.Df     RSS Df Sum of Sq      F   Pr(>F)
# 1     35 0.71028
# 2     34 0.68484  1  0.025447 1.5546 0.221237
# 3     33 0.54017  1  0.144668 8.8381 0.005475 **
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```




```
# Analysis of Variance Table
#
# Response: get(response)
#            Df  Sum Sq  Mean Sq F value   Pr(>F)
# pressure    1 0.07494 0.074940  4.5782 0.039867 *
# D           1 0.02545 0.025447  1.5546 0.221237
# pressure:D  1 0.14467 0.144668  8.8381 0.005475 **
# Residuals  33 0.54017 0.016369
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
#  ANOVA for E
#
# Analysis of Variance Table
#
# Model 1: get(response) ~ pressure
# Model 2: get(response) ~ pressure + D
# Model 3: get(response) ~ pressure + D + pressure * D
#   Res.Df     RSS Df Sum of Sq       F    Pr(>F)
# 1     36 115.627
# 2     35 115.538  1     0.089  0.0687    0.7949
# 3     34  44.121  1    71.418 55.0356 1.327e-08 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```




```
# Analysis of Variance Table
#
# Response: get(response)
#             Df Sum Sq Mean Sq F value    Pr(>F)
# pressure     1  0.012   0.012  0.0094    0.9235
# D            1  0.089   0.089  0.0687    0.7949
# pressure:D   1 71.418  71.418 55.0356 1.327e-08 ***
# Residuals   34 44.121   1.298
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
#  ANOVA for GH2O
#
# Analysis of Variance Table
#
# Model 1: get(response) ~ pressure
# Model 2: get(response) ~ pressure + D
# Model 3: get(response) ~ pressure + D + pressure * D
#   Res.Df    RSS Df Sum of Sq       F    Pr(>F)
# 1     36 157627
# 2     35 157091  1       535  0.2524    0.6186
# 3     34  72106  1     84985 40.0729 3.218e-07 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```




```
# Analysis of Variance Table
#
# Response: get(response)
#            Df Sum Sq Mean Sq F value    Pr(>F)
# pressure    1  26362   26362 12.4302  0.001231 **
# D           1    535     535  0.2524  0.618640
# pressure:D  1  84985   84985 40.0729 3.218e-07 ***
# Residuals  34  72106    2121
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3.6.2 Plots - Treatment-level

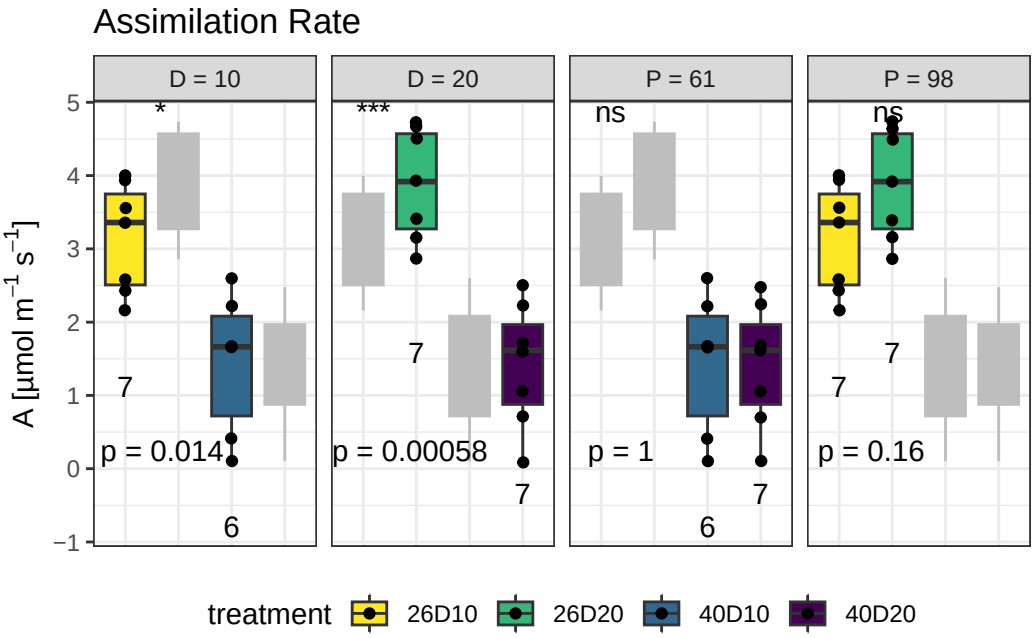


Figure 15: Carbon assimilation rate [mmol $m^{-1}s^{-1}$] by treatment, comparing across pressures and humidities. Significance label of the comparison above each facet, p-value below, and number of replicates (n) below each highlighted boxplot.

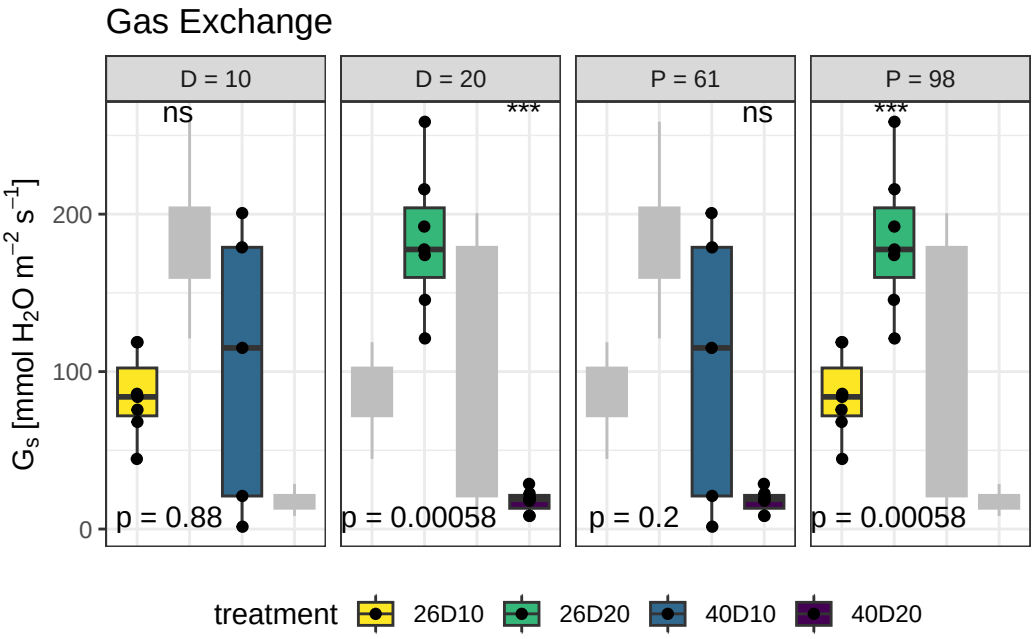


Figure 16: Stomatal conductance [mmol $m^{-1}s^{-1}$] by treatment, comparing across pressures and humidities. Significance label of the comparison above each facet, p-value below, and number of replicates (n) below each highlighted boxplot.

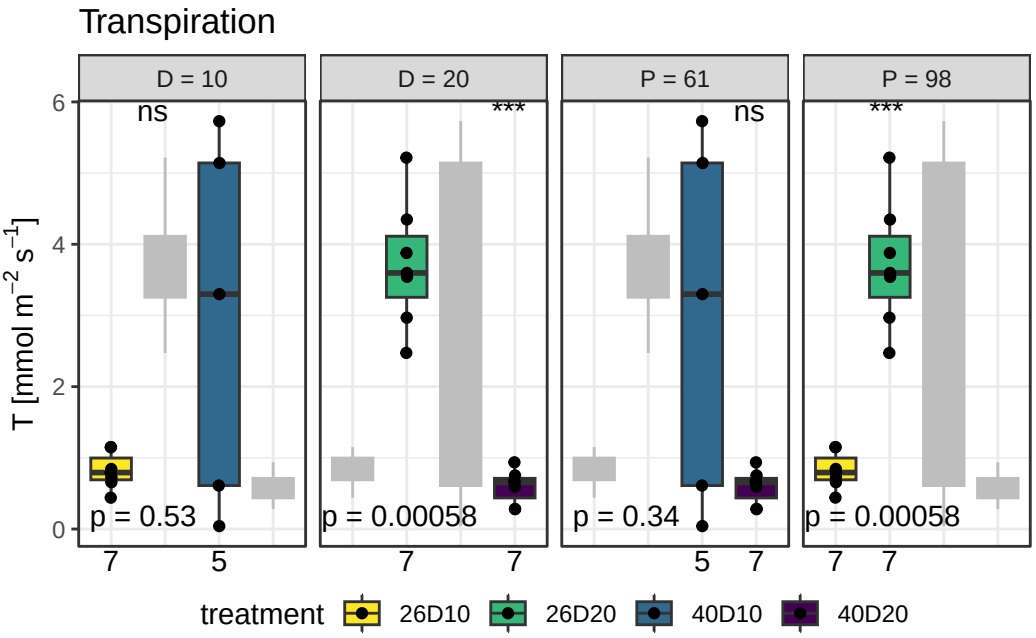


Figure 17: Transpiration rate [mmol $m^{-1}s^{-1}$] by treatment, comparing across pressures and humidities. Significance label of the comparison above each facet, p-value below, and number of replicates (n) below each highlighted boxplot.

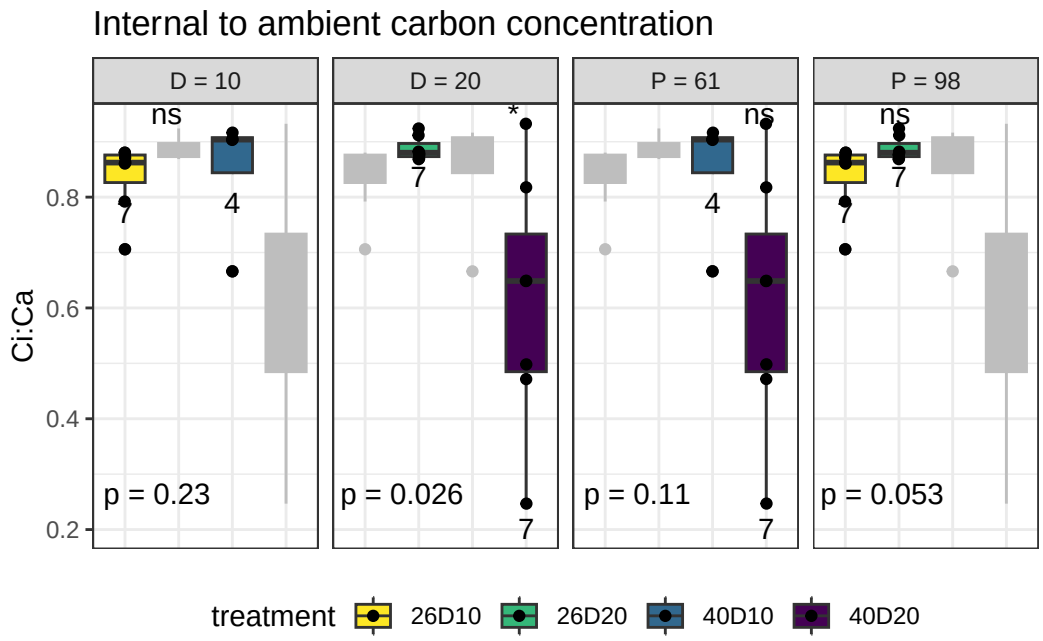


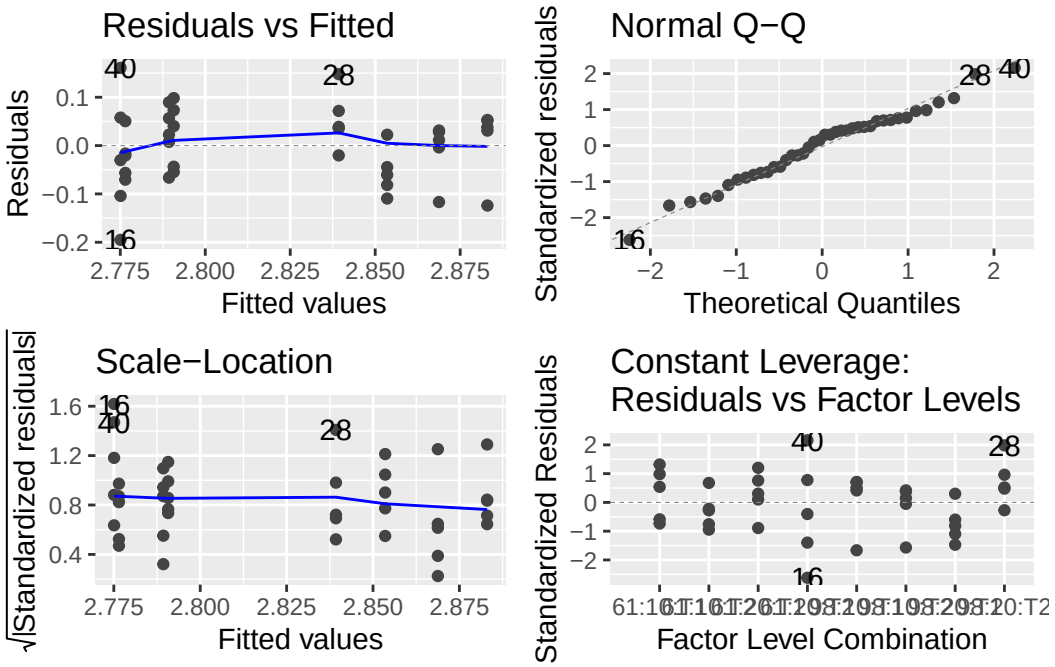
Figure 18: Ci:Ca by treatment, comparing across pressures and humidities. Significance label of the comparison above each facet, p-value below, and number of replicates (n) below each highlighted boxplot.

3.7 Pigments

3.7.1 General ANOVAs for Pigments

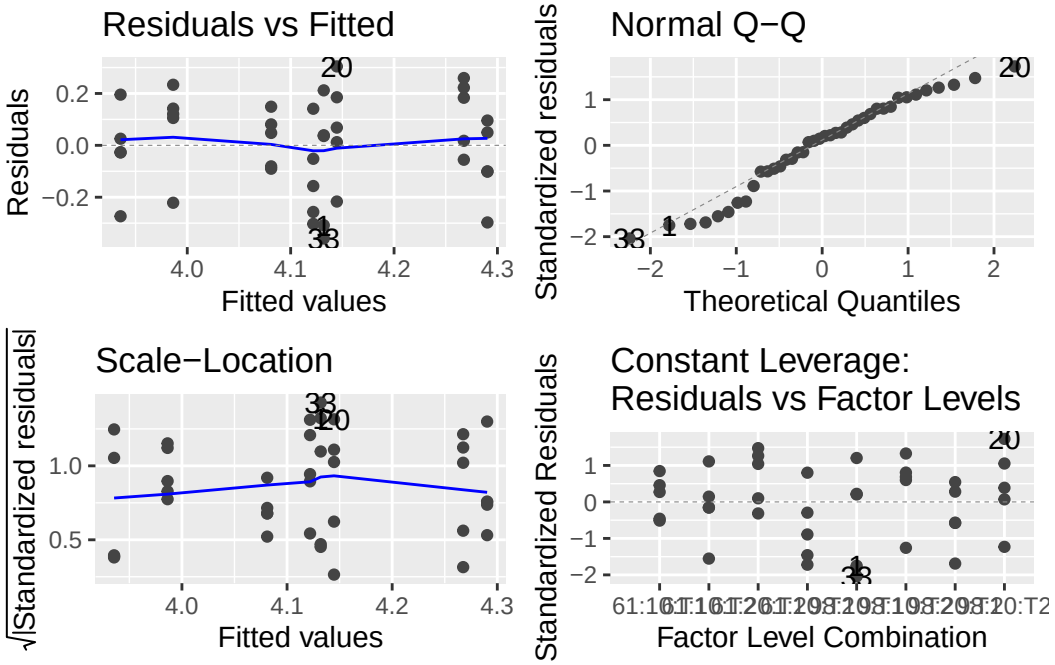
```
#
#  ANOVA for chlab
#
# Analysis of Variance Table
#
# Model 1: get(response) ~ time
# Model 2: get(response) ~ pressure + time
# Model 3: get(response) ~ pressure + D + time
# Model 4: get(response) ~ pressure + D + pressure * D + time
#   Res.Df     RSS Df Sum of Sq      F   Pr(>F)
# 1     38 0.28760
# 2     37 0.22637  1  0.061231 9.6530 0.003739 **
# 3     36 0.22398  1  0.002387 0.3763 0.543549
# 4     35 0.22201  1  0.001974 0.3112 0.580493
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# Analysis of Variance Table
#
# Model 1: get(response) ~ time
```

```
# Model 2: get(response) ~ D + time
# Model 3: get(response) ~ pressure + D + time
# Model 4: get(response) ~ pressure + D + pressure * D + time
#   Res.Df     RSS Df Sum of Sq      F   Pr(>F)
# 1     38 0.28760
# 2     37 0.28521  1  0.002387 0.3763 0.543549
# 3     36 0.22398  1  0.061231 9.6530 0.003739 **
# 4     35 0.22201  1  0.001974 0.3112 0.580493
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



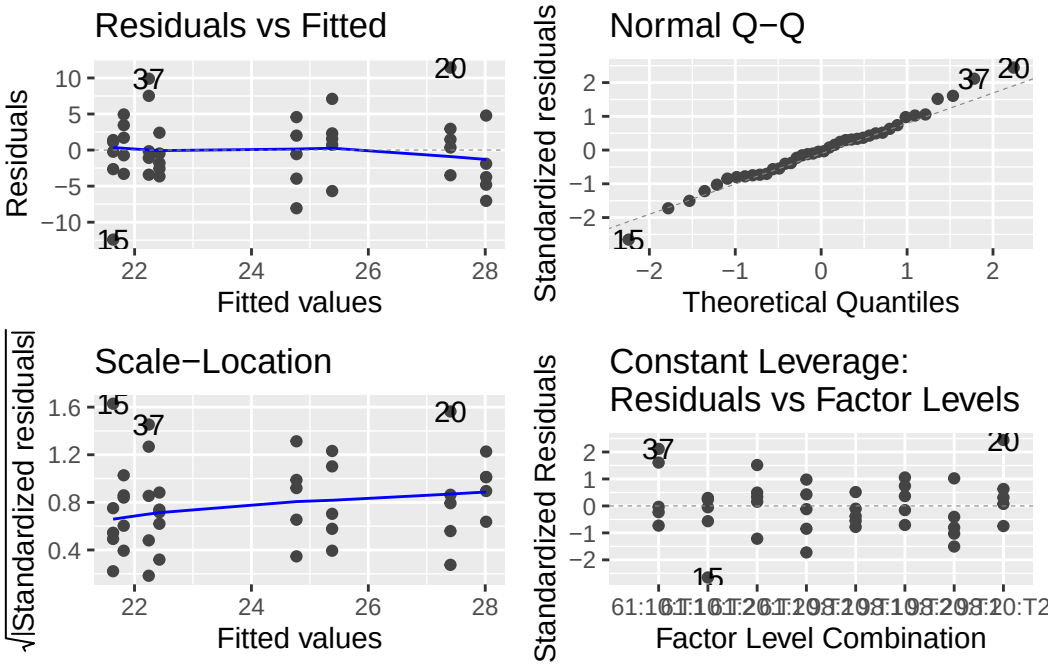

```
#
#  ANOVA for chlcar
#
# Analysis of Variance Table
#
# Model 1: get(response) ~ time
# Model 2: get(response) ~ pressure + time
# Model 3: get(response) ~ pressure + D + time
# Model 4: get(response) ~ pressure + D + pressure * D + time
#   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
# 1     38 1.5522
# 2     37 1.5386  1  0.013579 0.3834 0.539794
# 3     36 1.2416  1  0.297045 8.3871 0.006473 **
# 4     35 1.2396  1  0.001974 0.0557 0.814742
# ---
```

```
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# Analysis of Variance Table
#
# Model 1: get(response) ~ time
# Model 2: get(response) ~ D + time
# Model 3: get(response) ~ pressure + D + time
# Model 4: get(response) ~ pressure + D + pressure * D + time
#   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
# 1     38 1.5522
# 2     37 1.2551  1  0.297045 8.3871 0.006473 **
# 3     36 1.2416  1  0.013579 0.3834 0.539794
# 4     35 1.2396  1  0.001974 0.0557 0.814742
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

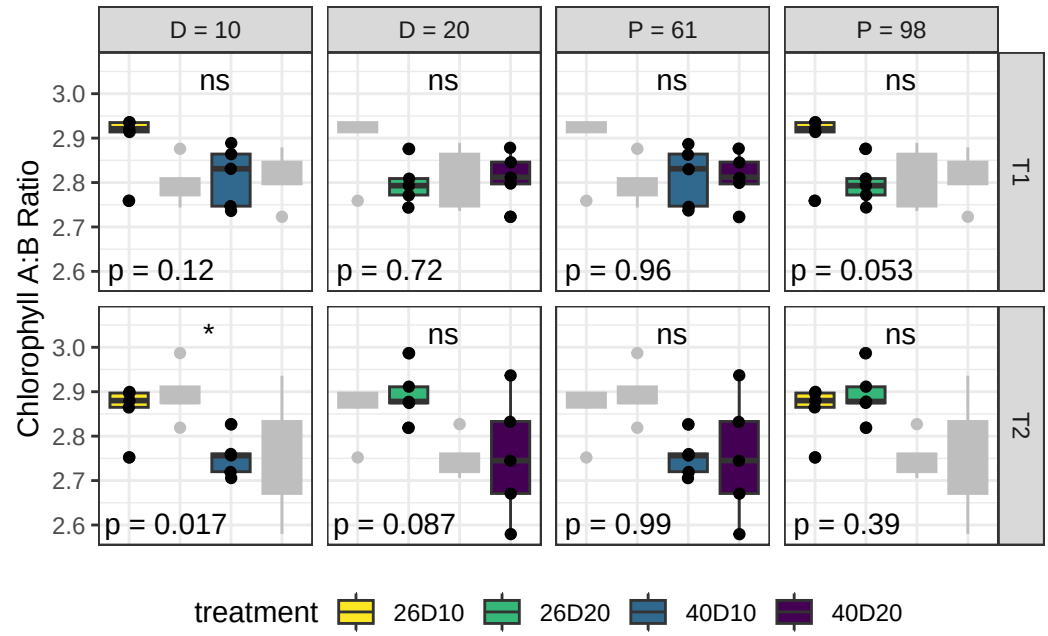



```
#
#  ANOVA for density
#
# Analysis of Variance Table
#
# Model 1: get(response) ~ time
# Model 2: get(response) ~ pressure + time
# Model 3: get(response) ~ pressure + D + time
# Model 4: get(response) ~ pressure + D + pressure * D + time
#   Res.Df     RSS Df Sum of Sq      F   Pr(>F)
# 1     38 1101.28
```

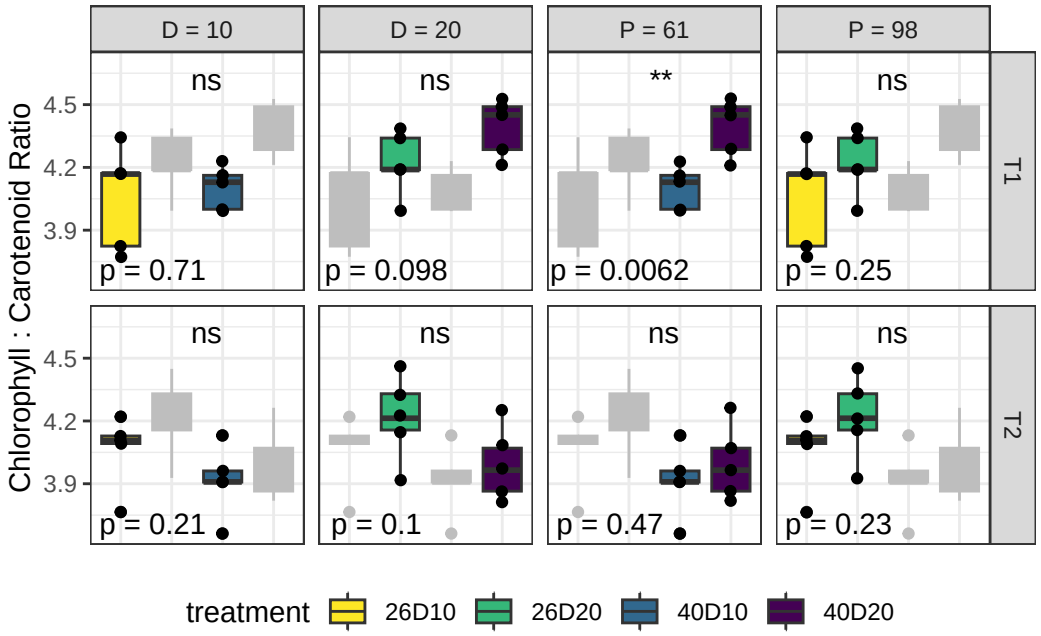
```
# 2     37 1081.44  1    19.833 0.7924 0.379443
# 3     36  891.04  1   190.407 7.6077 0.009177 **
# 4     35  875.99  1    15.042 0.6010 0.443403
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
# Analysis of Variance Table
#
# Model 1: get(response) ~ time
# Model 2: get(response) ~ D + time
# Model 3: get(response) ~ pressure + D + time
# Model 4: get(response) ~ pressure + D + pressure * D + time
#   Res.Df     RSS Df Sum of Sq      F   Pr(>F)
# 1     38 1101.28
# 2     37  910.87  1   190.407 7.6077 0.009177 **
# 3     36  891.04  1    19.833 0.7924 0.379443
# 4     35  875.99  1    15.042 0.6010 0.443403
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



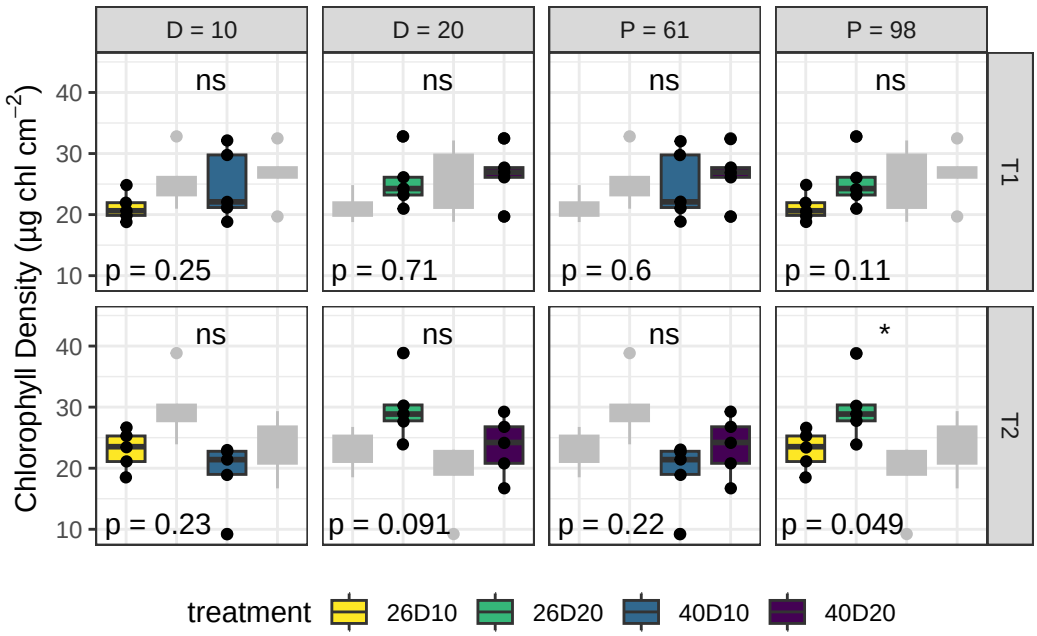
3.7.2 Chlorophyll A:B



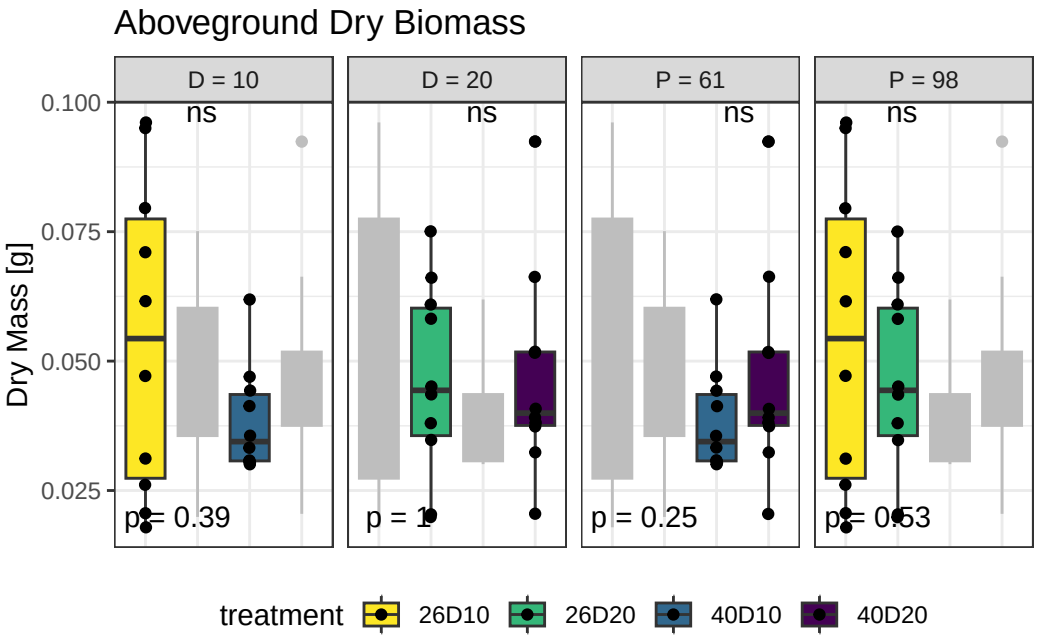
3.7.3 Chlorophyll : Carotenoid

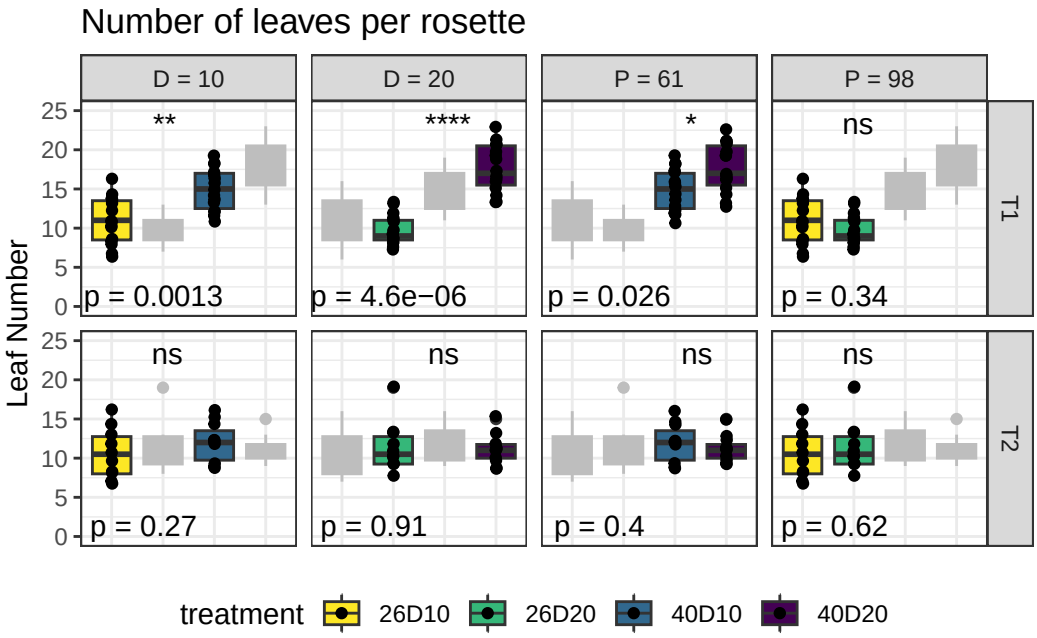


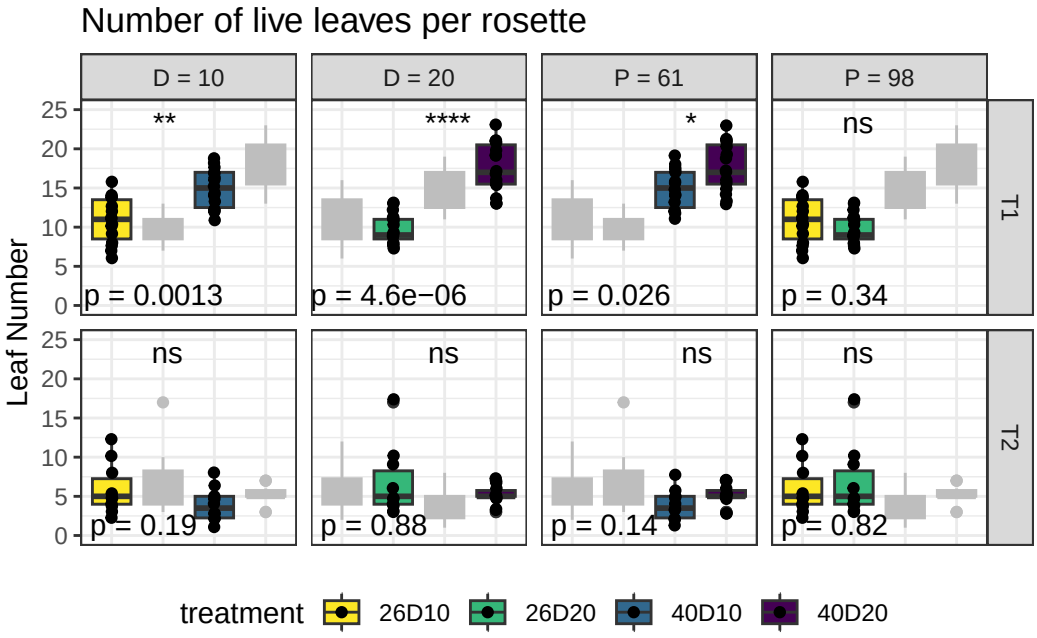
3.7.4 Chlorophyll Density

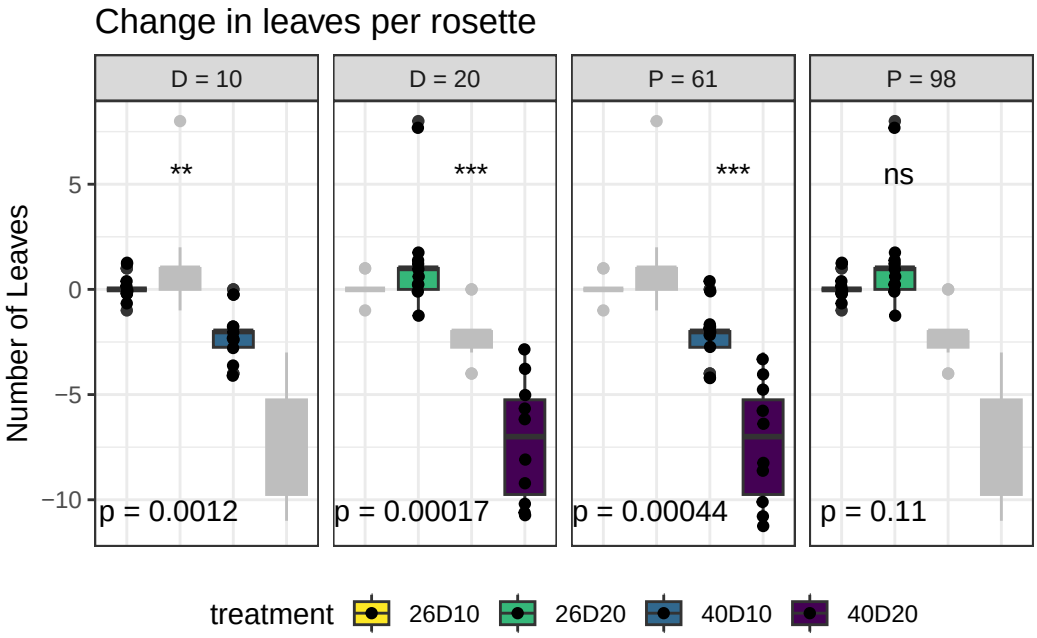


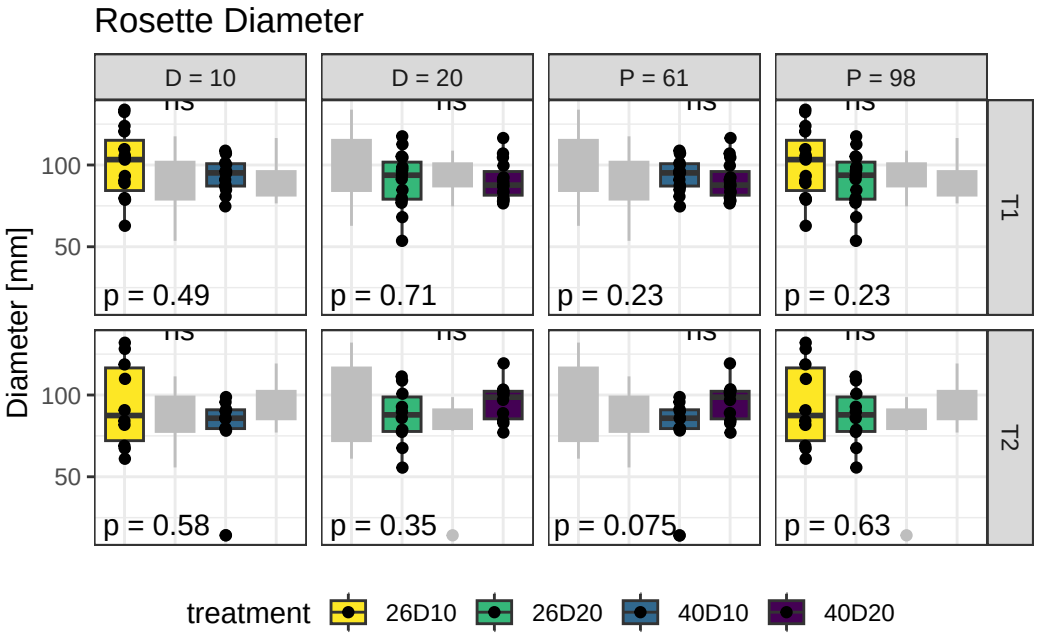
3.8 Biometric Data




Number of leaves per rosette
D = 10
D = 20
P = 61
P = 98
T1
T2
Leaf Number
**
****
*
ns
p = 0.0013
p = 4.6e−06
p = 0.026
p = 0.34
p = 0.27
p = 0.91
p = 0.4
p = 0.62
treatment
26D10
26D20
40D10
40D20


Number of live leaves per rosette
D = 10
D = 20
P = 61
P = 98
T1
T2
Leaf Number
**
****
*
ns
p = 0.0013
p = 4.6e−06
p = 0.026
p = 0.34
p = 0.19
p = 0.88
p = 0.14
p = 0.82
treatment
26D10
26D20
40D10
40D20


Change in leaves per rosette
D = 10
D = 20
P = 61
P = 98
Number of Leaves
5
0
−5
−10
**
***
***
ns
p = 0.0012
p = 0.00017
p = 0.00044
p = 0.11
treatment
26D10
26D20
40D10
40D20


Rosette Diameter
D = 10
D = 20
P = 61
P = 98
T1
T2
Diameter [mm]
100
50
ns
p = 0.49
p = 0.71
p = 0.23
p = 0.23
p = 0.58
p = 0.35
p = 0.075
p = 0.63
treatment
26D10
26D20
40D10
40D20